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A nexus assessment of Energy and Water in Rwanda

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Abstract

Rwanda is a small country in Central Africa with plans on 100% electrification and 60% renewable energy in the electricity mix by 2030. Today 31% of the country's population has access to electricity and hydropower yields approximately half (44%) of the country's electricity generation. The country's plans include an aggressive development plan, which aims to foster multi-sectoral development and elevate the country from poverty. Water is a critical resource, heavily exploited by both the power and agriculture sector. This thesis intends to investigate the role of hydropower in Rwanda's electrification plans, identify critical points in the use of water and explore the possible effects of climate change. Two energy modelling tools were used, namely OnSSET and OSeMOSYS. Water balance equations were developed manually based water demand data from the Rwandan Ministry of Natural Resources and three climate scenarios based on data from KNMI Climate Explorer. The results show that hydropower's share is expected to vary between 7 - 47% in the electricity mix by 2030, with its penetration highly depending on both electricity demand and climate scenario selected. It was also concluded that water availability will not be able to sufficiently cover the expected water demand in the country after 2026. Hence, Rwanda's aggressive hydropower expansion plans might need to be reconsidered.

Sammanfattning

Rwanda är ett litet land i centrala Afrika med mål att elektrifiera hela landet samt att ha en elmix med 60% förnyelsebart år 2030. Idag har 31% tillgång till el och vattenkraft står för 44% av elmixen. Vatten är emellertid en begränsad resurs med ökande behov från inte minst jordbruket. Denna rapport utreder vattenkraftens roll i Rwandas elektrifieringsplaner och klimatförändringars påverkan på vattentillgång.

För att modellera elmixen har OnSSET och OSeMOSYS som är två modeller utvecklade på KTH dESA använts. Beräkningar på vattentillgänglighet har gjorts med hjälp av vattentillgång utifrån flera klimatscenarion samt prognoser av framtida vattenbehov från Ministry of Natural Resources i Rwanda.

De framtagna resultaten visade att vattenkraften år 2030 bidrar med 7 - 47% till elmixen beroende på klimatscenario och elbehov. Denna variation beror dock framförallt på skillnaderna i de årliga variationerna i nederbördsmonster mellan olika klimatscenarion snarare än minskad total vattentillgång pga klimatförändringar.

Oberoende av klimatförändringarna kommer vattentillgången inte kunna täcka det uppskattade vattenbehovet och de framtida vattenplanerna kan därför behöva omprövas.

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Acronyms and Abbreviations

AEZ - Agro-Ecological Zones

CLEW - Climate, Land, Energy and Water strategies to navigate the nexus

DRC - The Democratic Republic of Congo

FAO - Food and Agriculture Organization of the United Nations

GDP - Gross Domestic Product

KTH dESA - division of Energy System Analysis at Royal Institute of Technology

LEAP - Long-range Energy Alternatives Planning System

OnSSET - Open Source Spatial Electrification Tool

OSeMOSYS - Open Source Energy Modelling System

PV - Photovoltaic technology

RCP - Representative Concentration Pathways

SDGs - United Nations' Sustainable Development Goals

UN - United Nations

Vision 2020 - A Government development program in Rwanda, launched in 2000 by Rwandan president Paul Kagame

WEAP – Water Evaluation and Planning System

Acknowledgement

We would like to take this opportunity to show gratitude to some key persons for this project.

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In Rwanda we would like to thank Aimable Habinshuti, Planning Engineer at Rwanda Energy Group for useful data on the energy system, Francois Tetero at Rwanda Water and Forestry Authority for useful data on future water demand and Hanson Micomyiza, Irrigation Specialist at Rwanda Agriculture Board for useful irrigation plans.

At last we would like to thank Åforsk for the fundings that made this study possible.

1. Introduction

Rwanda is a small landlocked country in Central Africa. With an area of approximately 26 thousands square kilometres it borders with Tanzania in east, Burundi in south, Democratic Republic of Congo (DRC) in west and Uganda in north. It is the most densely populated country in Africa with a population growth rate of 2.53% per year (World Factbook, 2016). Only 31% of the population has access to electricity, 76.1% to drinking water and it is estimated that approximately 16% is living below the extreme poverty line.¹ It is also estimated that about 74.5% of the land is used for agriculture, which accounts to approximately 33% of GDP (World Bank, 2015). Despite that, food production is inadequate to cover the country's demand, hampering in various ways sustainable development and growth. The imminent expansion of agricultural activity poses a number of dilemmas in the country's development plans. Agriculture depends on water availability, which is also a critical component to other activities such residential, commercial and industrial use.

The energy sector is particularly affiliated with resource scarcity and environmental implications. In 2012 approximately 86% of the primary energy demand in Rwanda was covered by biomass, mostly from solid fuels such as firewood and charcoal, used in households for cooking and heating. About 10% of the energy demand was covered by imported petroleum products and only 4% of the demand was covered by electricity (UNIDO, 2016a).

Disaggregating the electricity sector, it is estimated that about 90% is on grid and 10% off grid. On grid 38% derives from smaller scale hydropower (0.1 to 28 MW), 28% from methane gas, about 19% from diesel generators and the rest from imports, solar and peat (Habinshuti, 2017). Expanding access to electricity (Sustainable Development Goal (SDG) 7²) over the next years calls however for a significant increase in the country's power capacity. In fact, Rwanda has committed to a number of long term development targets defined under the Vision 2020. One of the goals envisions the increase of the electricity generation as well as the share of renewables in the country's generation mix. The target aims to achieve at least 60% of renewable energy in the electricity mix by 2030. Additionally, whilst today only 31% of the population has access to electricity, the target is that by 2030 the country will have reached an electrification rate of 100% (Habinshuti, 2017; Ministry of Infrastructure, 2016).

Hydropower is expected to have a significant contribution in the next years. Currently, the country has an installed capacity of 98.5 MW but the total estimated potential reaches 400 MW according to Rwanda's Ministry of Infrastructure. In addition, significant natural gas reserves have been identified in Lake Kivu, located in the west part of the country. The country's plentiful solar potential is also expected to contribute to electrification of rural areas, especially in the form of off-grid systems where connection to the national grid isn't a financially viable solution (Ministry of Infrastructure, 2015).

Like many other countries in Sub.-Saharan Africa, Rwanda is already suffering from natural hazards and with climate change occurrence, the risk for extreme weather conditions (such as long droughts and floods) increases even more. Additionally, Rwanda is one of the most vulnerable countries in the

¹ The daily income does not surpass 1,25 USD per day (World Bank, 2015)

² Sustainable Development Goal 7: Ensure access to affordable, reliable, sustainable and modern energy for all. The goal is one of 17 global goals set to 2030 by the United Nations (UNIDO, 2016a).

world, since climate change might significantly affect agriculture which as explained above is a sector that already receives a lot of pressure (Pauleit et al., 2015). The Food and Agriculture Organization of the United Nations (FAO) estimates that Rwanda's GDP will be reduced by 4.5% until 2030 due to climate change impact on the country's resources and mostly due to the irrigation changes affecting directly agriculture (FAO, 2016a).

2. Scope and objectives

The vulnerability to climate change in addition to Rwanda being so densely populated, induces significant pressure in the agricultural activities which should systematically expand in order to be able to cover food demand in the future (The World Factbook, 2016). Agriculture is also a sector with a high water demand and therefore the risk of a conflict between agriculture and the expansion of hydropower is evident. Thus, it is very important to include a water system analysis when designing pathways for the development of the future energy system in the country. Therefore, the questions this study aims to answer are:

- What is the role of hydropower in Rwanda's electrification plans?
- How will climate change affect the water availability and use in the country over the next 23 years?

The chosen time period was 2017-2040 to include Rwanda's water demand projections until 2040 and the targets of 100% access to electricity and 60% renewable energy in the electricity mix by 2030 (Ministry of Infrastructure, 2016; Ministry of Natural resources & Rwanda National resources, 2015).

2.1 Study boundaries

This study has only studied water availability and use in agriculture and the power sector. Other uses such as clean water for drinking and sanitation were not taken into consideration. Imports and exports of electricity were also not taken into account, even though the ability to exchange electricity with neighboring countries has a big impact on the optimal generation mix. Bilateral water use within neighboring countries as well as cross border water flow arrangements is yet another factor not included in this study.

3. Methodology

3.1 Literature review

Literature was reviewed in order to get a perception of the situation in Rwanda and to find a focus point for the study. The country's energy system, assets and shortcomings were of interest. Conflicts and possible synergies between resources and different sectors were also sought³. The focus point for

³ Sources being used were The World Factbook, government of Rwanda, United Nations Industrial Development Organization (UNIDO), FAO

the research was formed and the next part of the literature study involved learning the modelling tools OnSSET and OSeMOSYS (explained below) and investigating how a possible link between them is possible.

3.2 Modelling tools for sustainable development

On their seventieth anniversary in New York 2015 the United Nations set 17 Sustainable Development Goals (SDGs) in order to solve the environmental, economic, social and political challenges on both local and global level. The goals are set to 2030 and serve as a continuation of the earlier Millennium Goals from 2000 to 2015. One of the goals refers explicitly to Climate action (UNIDO, 2016b) examining in particular three resources that on one hand have a big impact on climate and on the other hand are vulnerable to climate change; namely land, energy and water. The resilience of these resources is crucial for a country to function and develop. They are thoroughly connected and also impact each other. A good way to estimate and adapt to climate change and its effects is therefore through an integrated analysis of climate change, land use, energy and water use.

3.2.1 CLEW

CLEW is an integration of different models investigating land, energy and water use developments with climate change factors taken in consideration. CLEW assessments usually include modelling with using LEAP⁴ (Long-range Energy Alternatives Planning System) or OSeMOSYS⁵ (Open Source Energy Modelling System) for energy, WEAP⁶ (Water Evaluation and Planning System) for water use and AEZ⁷ (Agro-Ecological Zones) for land use. The models can be connected through similarities between them. For example, hydropower will affect the water use and so will water needs for agriculture. Iterations are made between the models for calibration. Climate scenarios are also included in the integrations through e.g. rainfall patterns that will affect the water use (Howells et al. 2013).

3.2.2 OnSSET and OSeMOSYS

Two modelling tools which constitute a part of a CLEWs assessment are Open Source Spatial Electrification Tool (OnSSET⁸) and Open Source Energy Modelling System (OSeMOSYS). OnSSET compares different on grid and off grid technologies to find the most cost-efficient way to electrify all households until 2030 and therefore provides helpful guidelines for achieving the SDG 7 whilst OSeMOSYS finds the most cost efficient energy mix. This report presents six different scenarios for Rwanda's energy system created from integrations between these two models.

3.2.3 Representative Concentration Pathways

Representative Concentration Pathways (RCPs) are four different climate scenarios developed by Intergovernmental Panel on Climate Change (IPCC) in 2014. They are used for climate modelling and

⁴ <https://www.energycommunity.org/leap>

⁵ www.osemosys.org/

⁶ www.weap21.org

⁷ <http://www.gaez.iiasa.ac.at/>

⁸ <http://www.onsset.org/>

the four projections represent different radiative forcing values in year 2100; 2.6 W/m², 4.5 W/m², 6 W/m² and 8.5 W/m². These values correspond to a culmination of emissions in 2020, 2040, 2060 and no culmination respectively (van Vuuren et al., 2011).

3.3 Model Linking

OSeMOSYS was modeled to 2050 although the period regarded ends 2040. This because the model can do illogical investments when “time ends”. An attempt of integrating water demand into OSeMOSYS was performed in this study but in a rather simplified way (using MS Excel) due to time limitations.

After getting results in both OSeMOSYS and OnSSET the two models were linked. Capital investment, discounted operating cost and annual production by technology were outputs from OSeMOSYS that were calculated to the grid price as an input to OnSSET as seen in figure 1. The output from OnSSET, the demand from new settlements connected to the grid and from new installed solar stand-alone systems, were used as an input to OSeMOSYS. Updated results were retrieved with another optimal grid price and then 3 iterations were made until an integrated solution was found for every scenario.

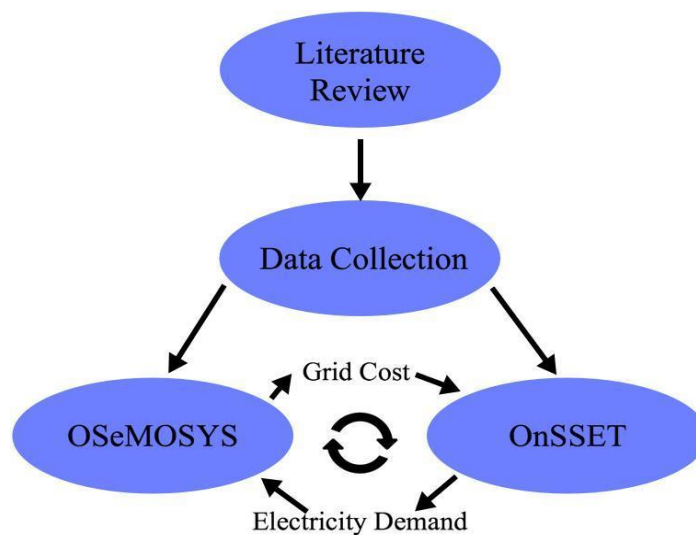


Figure 1. Process of the project

While OSeMOSYS divided the day into three splits (day, evening and night) OnSSET only considered one time split for the whole day. Since there wasn't a storage function for solar energy in OSeMOSYS this meant that solar energy couldn't be stored during the day and therefore not satisfy the evening demand. An adjustment in OSeMOSYS had to be done to be able to link the two models. Therefore, the entire demand from OnSSET (settlements with access to electricity after 2015) was placed during the day in OSeMOSYS.

3.3.1 Water Calculations

Water supply was calculated through collection of monthly precipitation (mm/m²) from Climate Explorer and multiplied with the total area of Rwanda (26 338 km²) (Koninklijk Nederlands Meteorologisch Instituut, 2017; The World Factbook, 2016). Water balance was then calculated in

Excel using the water supply and demand for each month. The model started in January of 2010, with the water demand subtracted from the water supply. For those months with a positive difference between supply and demand the surplus water was stored and added to the water supply of the next month. If the water supply was less than the demand at a given month, the following month was assigned to begin with an empty storage, yet with a new demand (even though the old one wasn't fulfilled). In OSeMOSYS the water supply for each year was set as a constraint of how much hydropower could be produced. To do this, the amount of water needed for a hydropower plant to produce one joule was estimated through an equation described in section 3.6.

3.4 Data Collection

Most data have been collected through interviews and available reports or provided by the division of Energy System Analysis at Royal Institute of Technology (KTH dESA) (as seen in Table 1 - for a more elaborate description please refer to Appendix B). Data regarding water supply (precipitation, humidity, temperature) were collected from Climate Explorer and for three different climate scenarios; RCP 2.6, RCP 6.0 and RCP 8.5 (Koninklijk Nederlands Meteorologisch Instituut, 2017).

Table 1. Data by Source

Source	Data
Aimable Habinshuti, Planning Engineer, Rwanda Energy Group	Population, number of households, electricity demand, access to electricity, grid losses, peak- and base demand, production by power plant, capacity by power plant
Francois Tetero, Director of Water Resources Regulation at Rwanda Water and Forestry Authority	Water demand for irrigation, urban and rural residency, livestock, industry, mining and quarries and fishing.
Hanson Micomyiza, Irrigation Specialist, Rwanda Agriculture Board	Irrigation plans
Climate Explorer, model GISS-E2-H p1	Precipitation, humidity, temperature 2010-2050
Availability Factor, Capacity Factor, Capital cost, Fixed cost, Efficiency, Operational Life, Electricity demand (Industry and residential electrified before 2015)	KTH dESA, The Electricity Model Base for Africa (TEMBA)
Grid Capacity Investment cost, Capital Cost for stand-alone and mini grid technologies	KTH dESA

3.5 Assumptions

Building up comprehensive energy models require a significant amount of input information. A number of assumptions were made for certain parameters for which a further sensitivity analysis was not a possibility due time and computational limitations. Here is a brief presentation of them:

- Discount rate was set at 8% in both OSeMOSYS and OnSSET.
- When missing annual data, linear interpolation was done between the given years (in the water demand data were only available for 2012, 2020, 2030 and 2040 and for some calculations the population growth and the urban ratio were used as a growth indicator). The interval between given years was considered so short that a more advanced interpolation method was not needed.
- Rwanda has potential peat capacity (15 MW is already in use and 80 MW is under construction) (Ministry of infrastructure, 2016). No data on costs, efficiency, emissions etc. for peat could be identified and therefore the average capital cost for power plants in Rwanda and the values of biomass from TEMBA⁹ were used.
- The categorization of the on-grid hydropower plants was made from the sizes of the current hydropower plants in Rwanda. For example, a large hydropower plant in this model might not be categorized as a large power plant in a bigger country but compared to other hydropower plants in Rwanda it was a large power plant. Accordingly, the hydropower plants were organized into three even groups; hydropower plants with a capacity less than 1 MW, hydropower plants with a capacity between 1-5 MW and hydropower plants with a capacity larger than 5 MW.
- In the calculation of the water amount needed for a hydropower plant to produce one joule the value for an average head was assumed to be 68 meters for the medium, small and off grid hydropower plants (< 5 MW) and 170 meters for the large hydropower plants (> 5 MW). No data for head was found but since water supply and hydropower production from earlier years exist an average head was calculated from those years.
- In the calculations for irrigation patterns for agriculture irrigation for hill sides and plains was used for the whole agricultural area even though other types like marshland has other irrigation needs. The patterns for hill sides and plains were more common and therefore more representative for the whole area.
- Irrigation plans are set for 2020 by Government of Rwanda. To model the demand to 2040 the potential irrigation area in the country was set as demand by 2050 and then linearly interpolated from 2020 to 2050.
- The climate data collected from Climate Explorer covered the area of coordinates S 0-4 and E 27,5 -30 meaning it covered not only Rwanda but also large parts of DRC, Burundi and Uganda. This because no more precise data for the location of Rwanda was found.
- The electricity demand used in the model was partly (the demand in the fuel EL3I) based on data from the TEMBA model. The demand in EL3R (electrified settlements 2015 - 2030) was made in

⁹ The Electricity Model Base for Africa (Taliotis et.al., 2016).

OnSSET with the target of 100% electricity access. After 2030 the demand in EL3R increase proportional to the population growth.

3.6 Calculations

Water supply was calculated from the formula $S = P - E - ET \pm SRO \pm GF$

where

- S is the water supply in cubic metres
- P is the precipitation in cubic metres
- E is the evaporation in cubic metres
- ET is the evapotranspiration in cubic metres
- SRO is the surface runoff in cubic metres
- GF is the groundwater flow in cubic metres

In this calculation surface runoff and groundwater were assumed negligible since the subtraction of evapotranspiration from precipitation was already negative for most months. For the calculation of evapotranspiration, the tool ETo Calculator¹⁰ developed by FAO was used with inputs of temperature (mean, min, max) and mean humidity.

The amount of water needed for a hydropower plant to produce one joule was calculated from the equation for potential energy $E = m \times g \times h$ where $m = \rho \times Q$ and

- E is the energy in joules
- m is the mass in kilograms
- ρ is the density of water in kilograms per cubic metres
- Q is the quantity of water in cubic metres
- g is the acceleration due to gravity in metres per second squared
- h is the head in metres

The parameters Base to Peak in OnSSET and the Specified Demand Profile in OSeMOSYS were calculated from data from Rwanda Energy Group. Peak demand was 120 MW and the lowest demand 50 MW (Habinshuti, 2017). The Base to Peak is the ratio between the lowest demand and the peak demand. The Specified Demand Profile in OSeMOSYS was more precise and a daily demand curve from the report *Energy Sector Strategic Plan* was used (Republic of Rwanda, 2015). In OSeMOSYS every day was split into three parts:

- Day (6 AM - 6 PM),
- Evening (6 PM - 10 PM) and

¹⁰ <http://www.fao.org/land-water/databases-and-software/eto-calculator/en/>

- Night (10 PM - 6 AM).

In the daily demand curve the Night part has the lowest demand and thus 50 MW. The demand is at its highest between 6 PM and 10 PM with the peak at 7 PM. From the curve in *Energy Sector Strategic Plan* an average demand in the Evening and Day part could be estimated. The average demand in the Evening part is about 100 MW and the Day average about 80 MW. These values were used to calculate the ratio between the different day splits. The different demands used in Specified Demand Profile can be seen in figure 2.

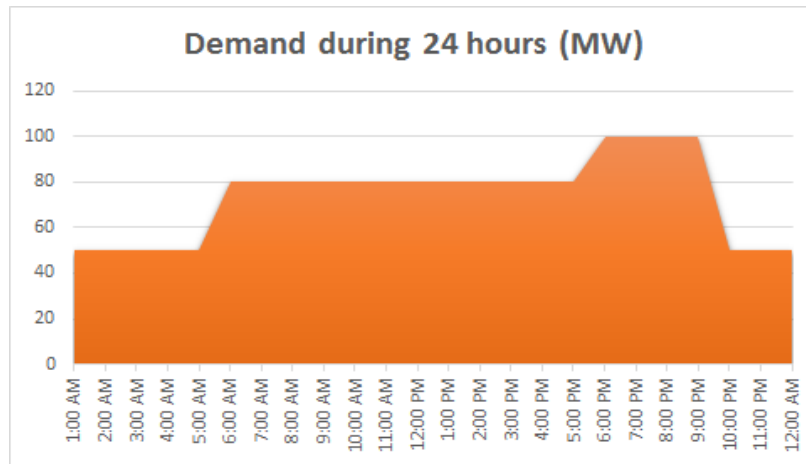


Figure 2. The variations in the demand used in OSeMOSYS

3.7 Sensitivity Analysis

The choice of discount rate can often have a big impact on the result in the optimization process of future systems. In this study, a discount rate of 8% was used but comparisons with 6 and 10% were also made. Hydropower which was the most important part of the electricity mix for this study didn't change noticeable with different discount rates and therefore the discount rate's impact on the results was negligible.

3.8 Scenario Formation

The scenarios used in this study were based on two energy demand levels (low and high) and three different climate scenarios. The low and high energy demands were created from the target of 100% electricity access. Of these 70% should have a consumption of UN's definition of Tier 2 (224 kWh/household/year) and 30% a consumption of Tier 3-5 (Tier 3: 696, Tier 4: 1800, Tier 5: 2195 kWh/household/year) (Ministry of infrastructure, 2016; UN DESA, 2012). The low demand is therefore 70% Tier 2 and 30% Tier 3 and the high demand 70% Tier 2 and 30% Tier 5. The climate scenarios used, namely RCP 2.6, RCP 6.0 and RCP 8.5, did represent the lowest and highest expected effect of climate change and a scenario in between.

4. Results

This chapter presents the results from modelling in OSeMOSYS and OnSSET together with the results from the water calculations in Excel. The results are arranged in the three different climate scenarios (RCP 2.6, RCP 6.0 and RCP 8.5) and a high and low electricity and water demand scenario. The high and low water demand varied only in the industry and mining and quarries demand which didn't affect the results significantly and therefore one figure represents both water demand scenarios for every climate scenario.

4.1 Energy

For an overview of the electricity mix from 2010 to 2040 see figures 3, 5 and 7. As can be seen the demand increases exponentially until 2030 due to the target of 100% electrification and stabilizes after 2030. As shown in figures 4, 6, 8 and 10 hydropower accounts for a big part of the electricity generated in all scenarios and especially in the low energy demand scenarios. Since the hydropower production is limited by water availability it can't be expanded from a low energy scenario to a high energy scenario. Together with solar energy hydropower contribute from 46% to 93% in the electricity generation mix. Therefore, the target of 60% share from renewable sources in the electricity generation mix is met all years, with an exception in year 2030 in the RCP 8.5 scenario because of low hydropower production due to water availability.

Nevertheless, hydropower production differs between years because of the variations in precipitation patterns and is hence considerably lower in some years. According to the findings of the study, climate change won't affect hydropower production significantly since the latter shows similar values in all climate scenarios tested. Various water supplies also result in "highs" and "lows" in hydropower production being compensated mainly by coal.

4.1.1 RCP 2.6

4.1.1.1 Low Demand

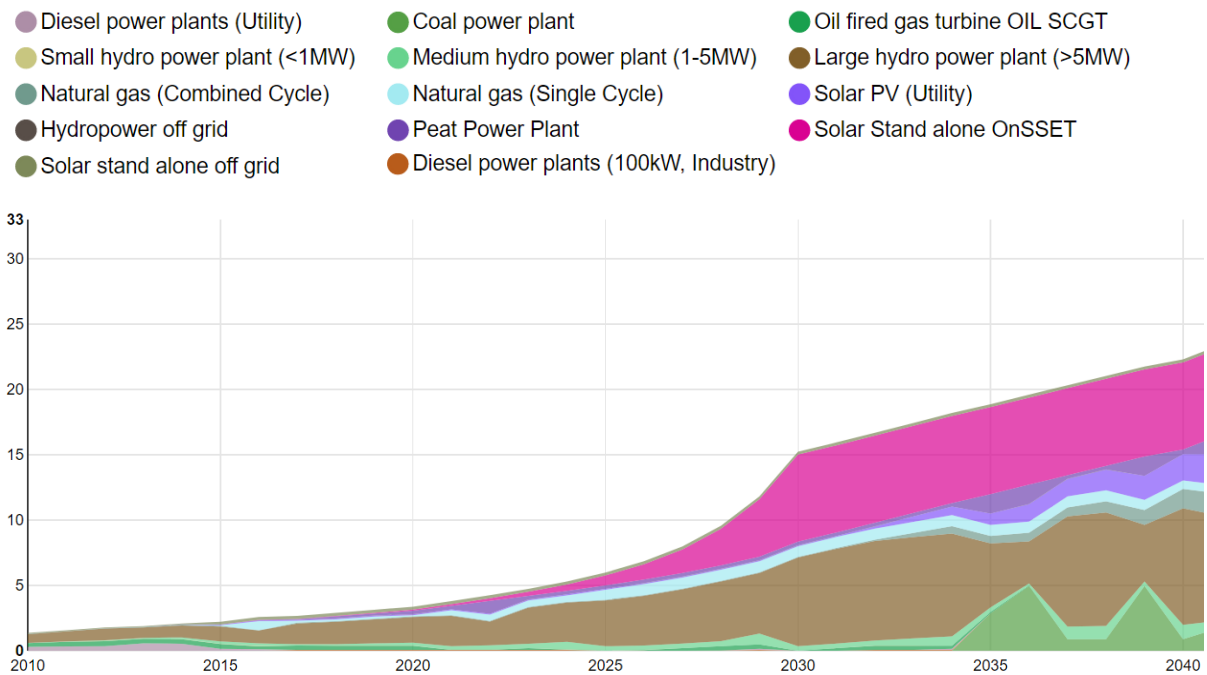


Figure 3. Energy Production by Technology Low Demand RCP 2.6 (PJ)

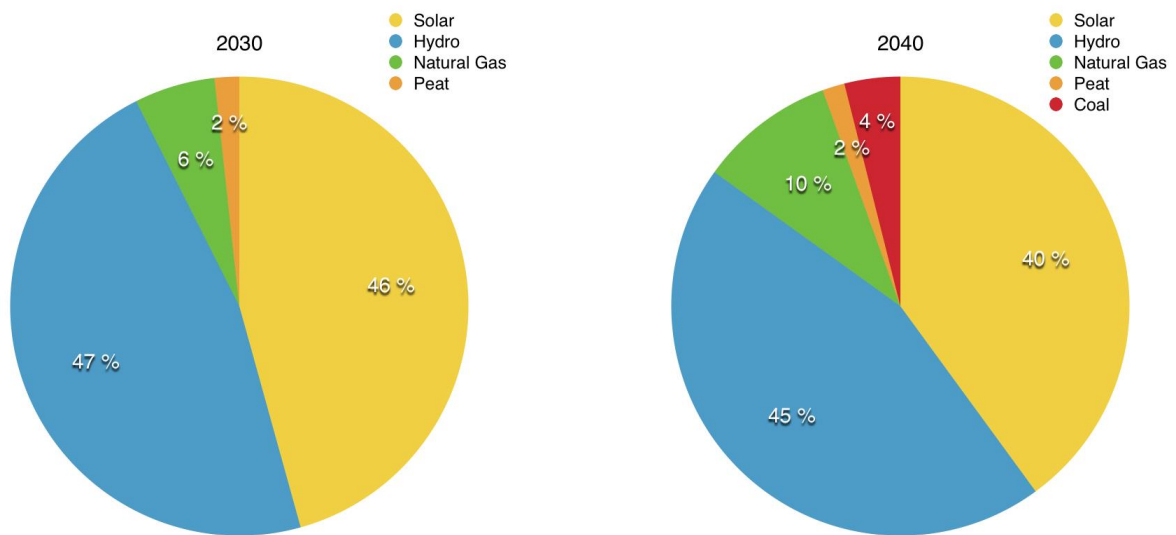


Figure 4. Production by Energy Source Low Demand RCP 2.6

4.1.1.2 High Demand

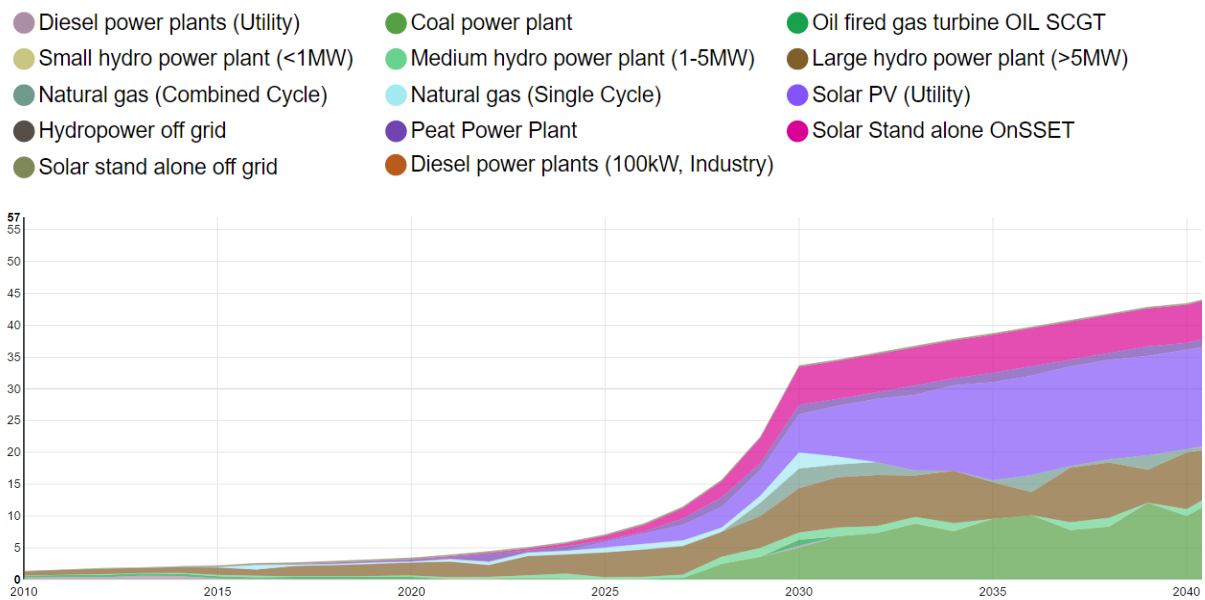


Figure 5. Energy Production by Technology High Demand RCP 2.6 (PJ)

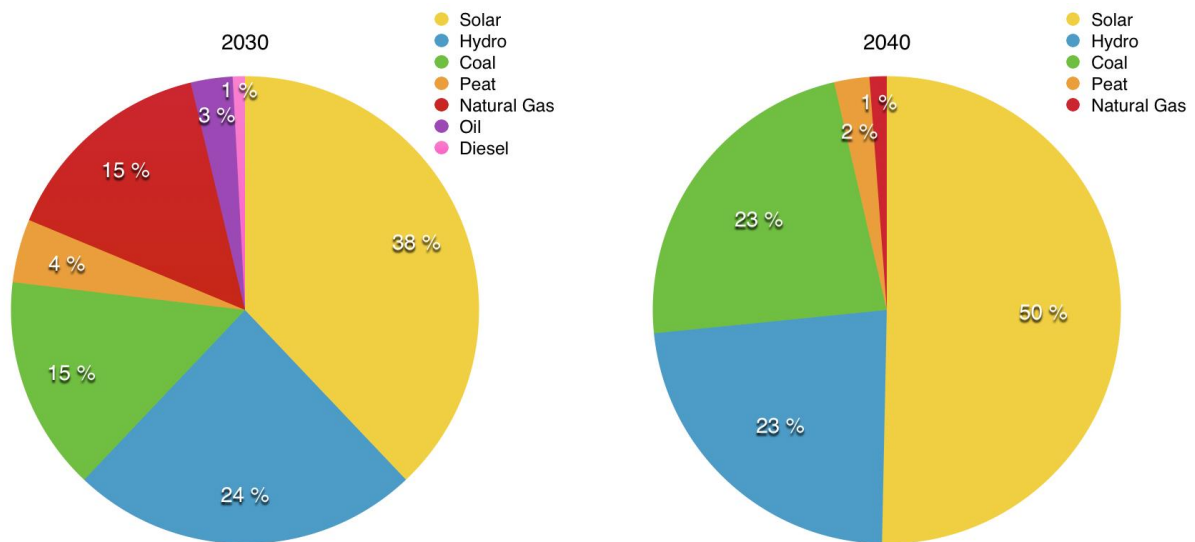


Figure 6. Production by Energy Source High Demand RCP 2.6

4.1.2 RCP 6.0

4.1.2.1 Low Demand

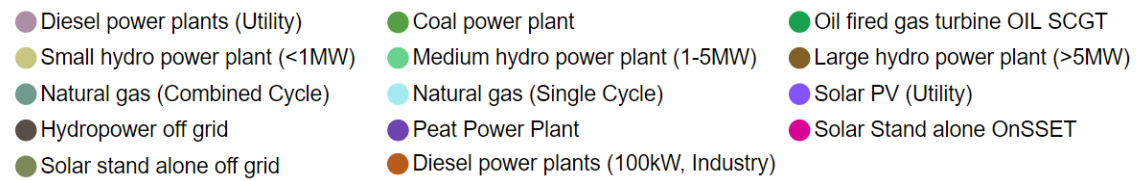


Figure 7. Energy Production by Technology Low Demand RCP 6.0 (PJ)

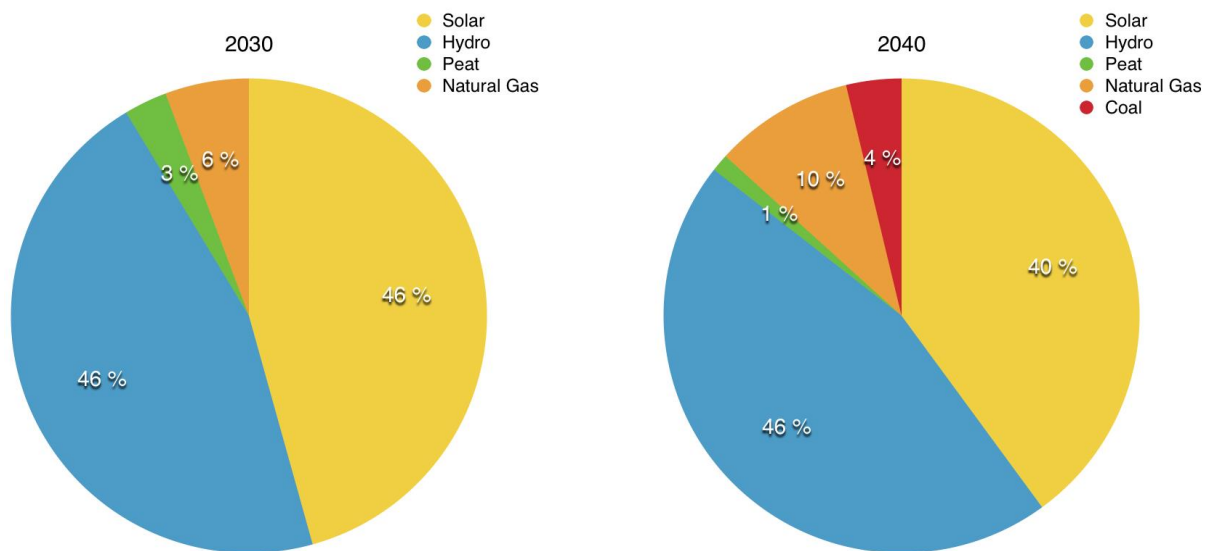


Figure 8. Production by Energy Source Low Demand RCP 6.0

4.1.2.2 High Demand

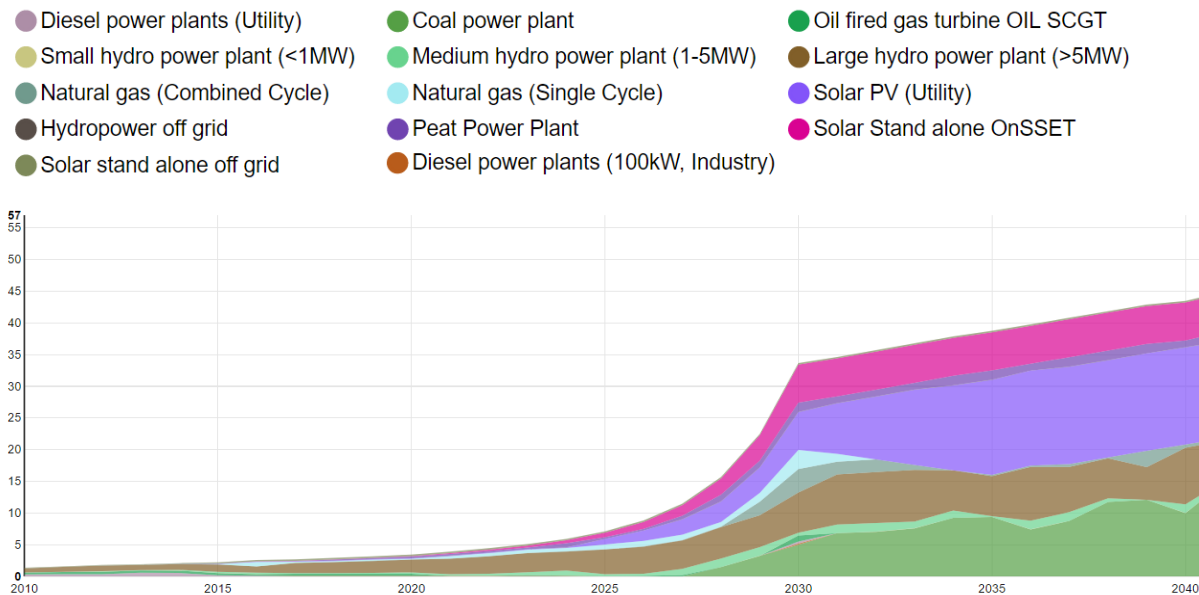


Figure 9. Energy Production by Technology High Demand RCP 6.0 (PJ)

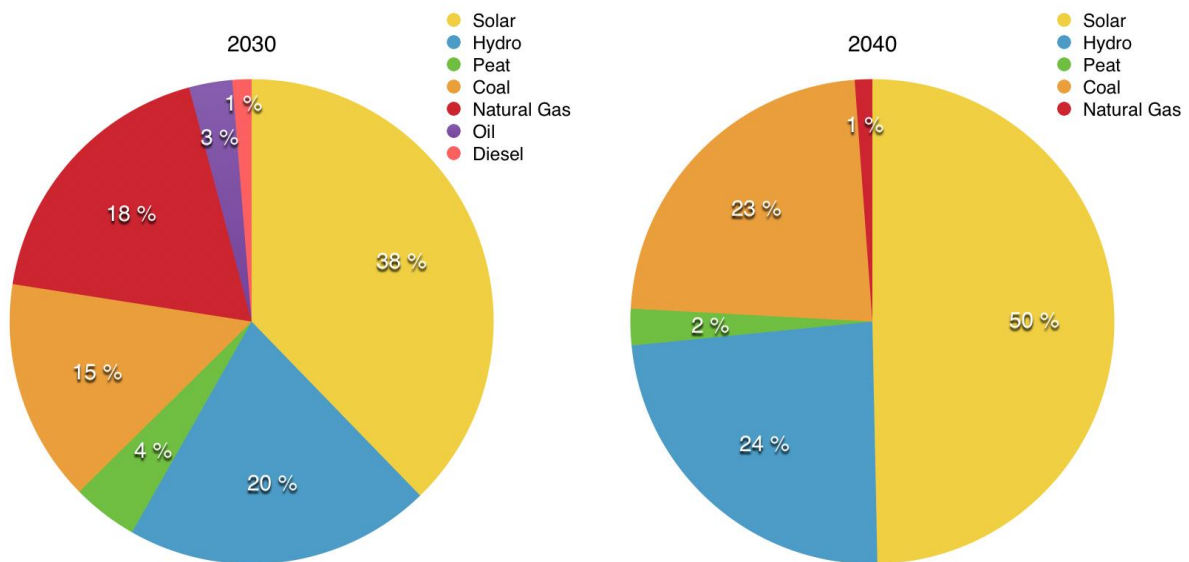


Figure 10. Production by Energy Source High Demand RCP 6.0

4.1.3 RCP 8.5

4.1.3.1 Low Demand

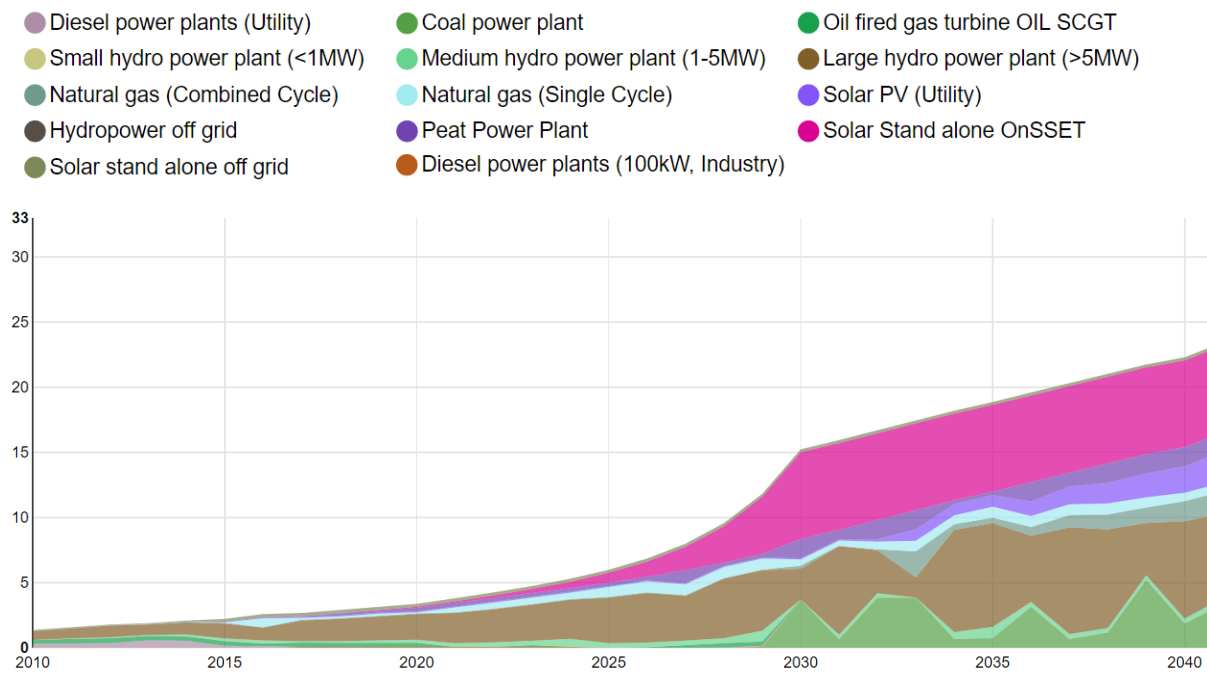


Figure 11. Energy Production by Technology Low Demand RCP 8.5 (PJ)

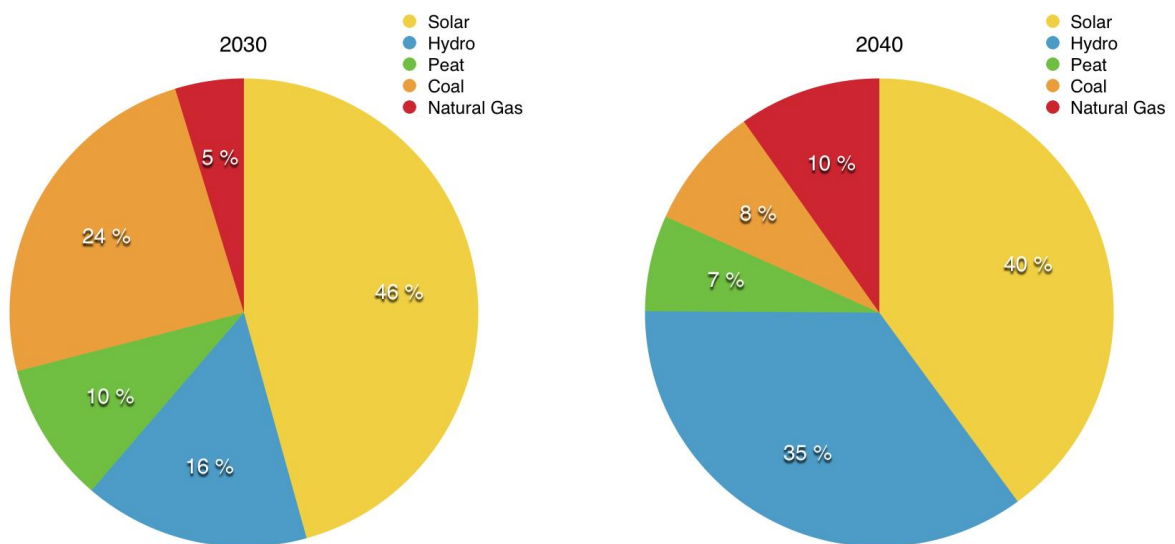
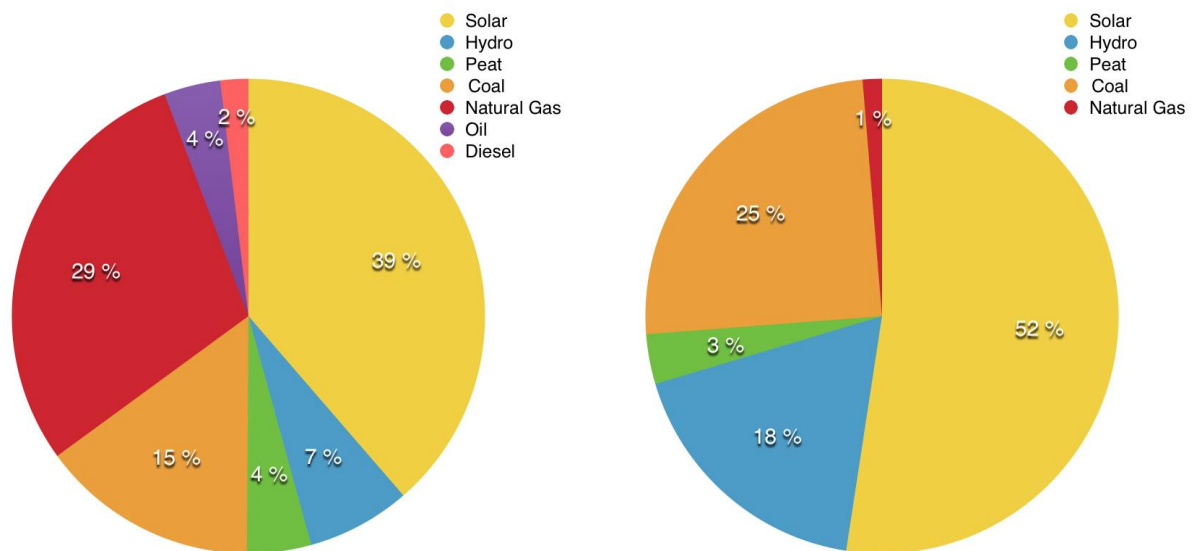
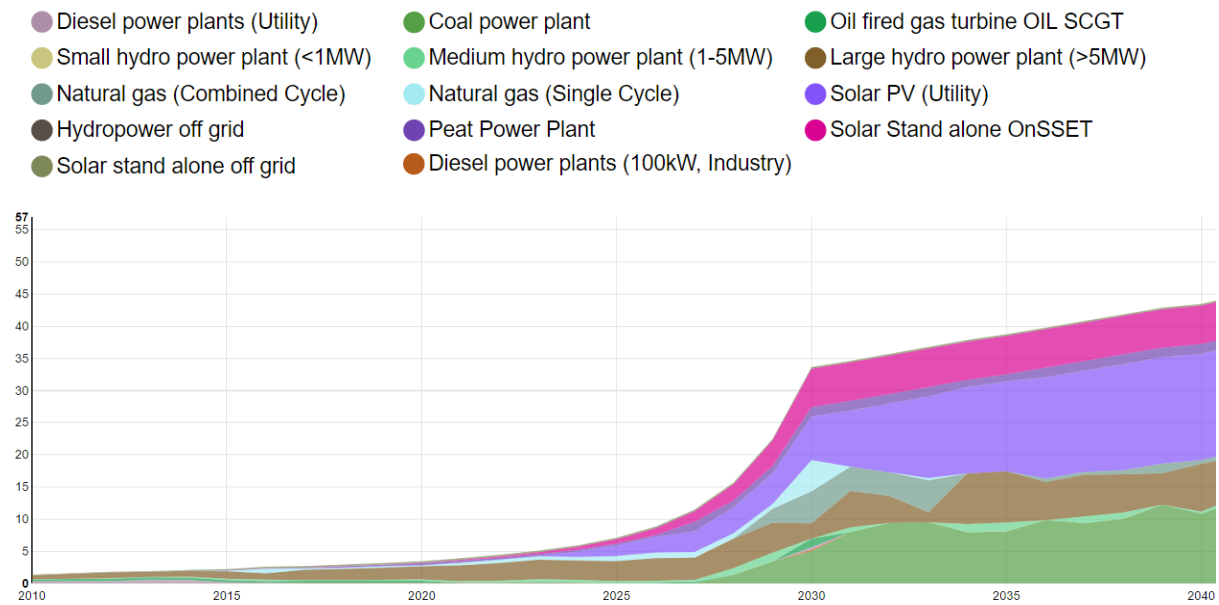


Figure 12. Production by Energy Source Low Demand RCP 8.5

4.1.3.2 High Demand



4.2 Water

The following water figures (Figure 15, 16, 17) show that the water supply isn't high enough to cover the demand for every month. With a storage function in lakes and dams it is however possible to use the surplus from rainy months to the demand in the dry seasons. But this will only help for a part of the modelling period. All models reach a point when demand can't be met despite storage. The first time is between 2025 - 2027 and after that it basically happens in every dry season until the end of the modelling period. The RCP 2.6 and RCP 6.0 scenario are very similar and have almost the same amount of months with unmet demand. The RCP 8.5 scenario on the other hand has almost one extra month per year of unmet demand than the other two. For additional details refer to Appendix B.

4.2.1 RCP 2.6

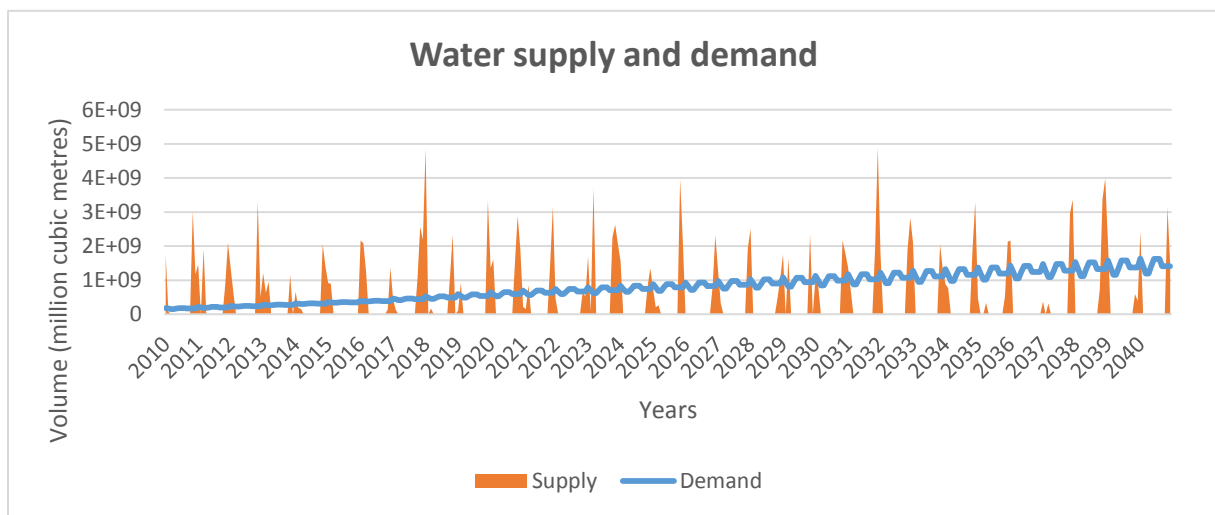


Figure 15. Water Availability RCP 2.6 (million m³)

4.2.2 RCP 6.0

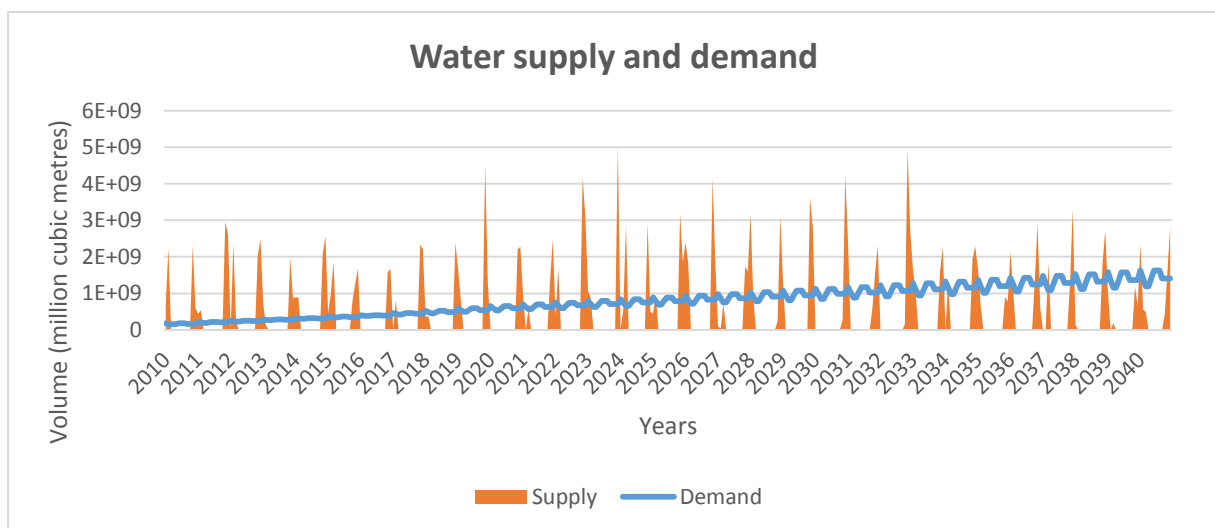


Figure 16. Water Availability RCP 6.0 (million m³)

4.2.3 RCP 8.5

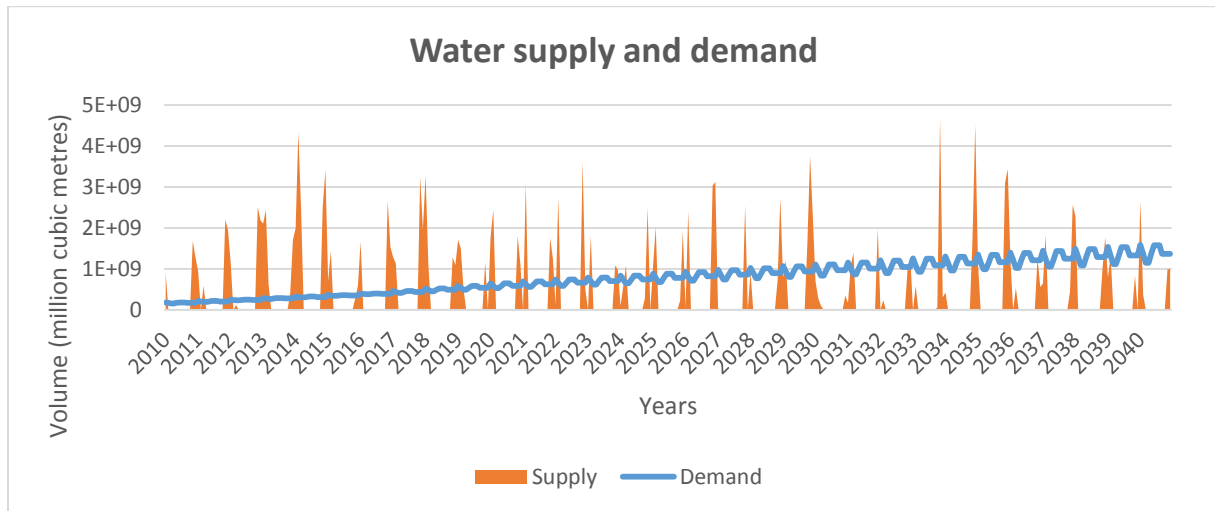


Figure 17. Water Availability RCP 8.5 (million m³)

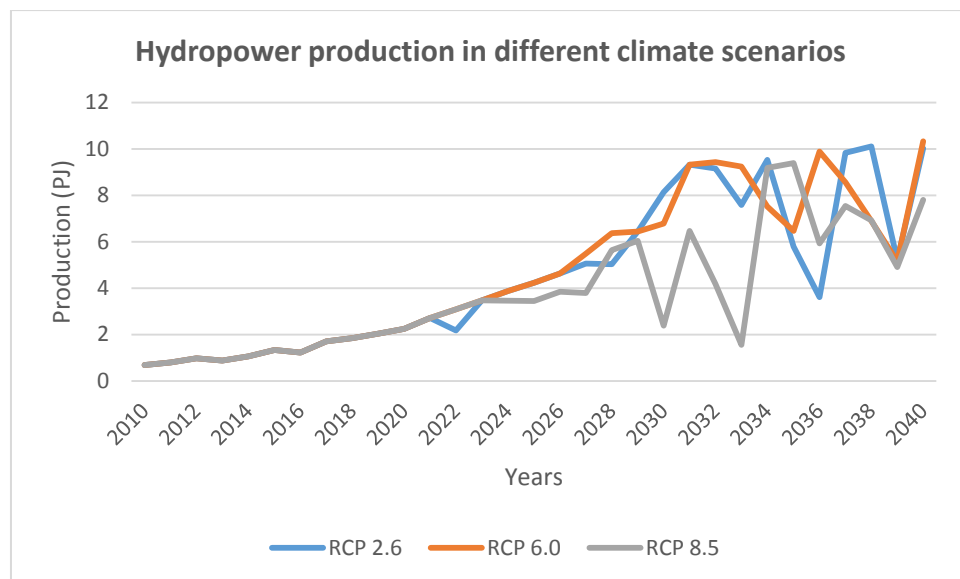


Figure 18. Hydropower variations due to different climate scenarios (PJ)

As seen in figure 18 the water availability will affect the hydropower production annually and result in some years with higher production and some years with a lower production. Even though the peaks and lows varies over time between the scenarios the different sums of the production especially in RCP 2.6 and RCP 6.0 are similar. However, the RCP 8.5 scenario has a significant lower production in 2029-2034 due to the water availability.

Discussion

According to the results of this study Rwanda's target of 60% renewable energy in the electricity mix is feasible for most years and climate scenarios. However, these results show the most cost-efficient electricity mix and although the solar and hydropower potential are maximized the target could be accomplished with less cost-efficient power production options such as wind or geothermal power. It is also arguable whether these results show the most cost-effective mixture when some input data was not found due to limitations in time. For example, were the costs regarding biomass used for peat as well.

Different climate scenarios didn't have noticeably impact on the hydropower production except from a few years in the model. It is however important to consider an elaborate water management where the supply first is being used in a hydropower plant before for example irrigation to fully use this hydropower potential. The monthly variations are different in different scenarios but the total water supply in the modelling period doesn't change much due to climate change. Despite its low impact on water availability climate change can still have devastating effects such as ruined infrastructure due to floods.

Regardless climate changes the water supply will not be able to meet the demand after 2026. Therefore, some considerations need to be done about the future water plans. The forecasts of the demand are hard to change but the planned expansion of irrigation, water for livestock and water used for mining and quarries aren't set. There might be other possibilities to meet the food demand through a more efficient agriculture and import from other countries. For instance, one observation from the study in Rwanda is the large number of cows. 1 kg beef needs 15 times more water than 1 kg pulses, roots or tubers (FAO, 2016b). According to the results Rwanda's government thus needs to choose between different water demands.

In addition, the water supplies presented in this study should be regarded as an approximation. Apart from the obvious that they are forecasts there are other factors that affect the results. Water losses weren't considered in the study and precipitation (mm/m^2) was collected from an area greater than Rwanda (S 0-4 and E 27.5 -30), thus not specific for the location of Rwanda.

The expansion plans for different power plants as well as 100% electrification by 2030 are ambitious targets and the electricity demand which is strongly correlated to these might therefore not look as presented in this report. However slower electrification rate doesn't mean that the increase in demand won't occur, only that it will be postponed.

Considering Rwanda's situation, rapidly developing with much development still ahead, there are great opportunities for a sustainable development avoiding paths taken by others. Today the planet is facing an unsustainable situation and industrial countries need to do major transformations in their habits and existing energy systems to eliminate their dependency on fossil fuels. Therefore, developing energy systems based on renewable energy could be a way to avoid large-scale changes and costs in the future.

Nevertheless, climate change is global and not national and therefore it is paramount that industrial countries commit to their responsibilities and make these transformations to avoid devastating effects, not least in vulnerable countries like Rwanda.

Conclusions and Recommendations for future work

The conclusions that can be made from the electricity model and the water calculations are that climate change will affect the water availability but not dramatically. Hydropower will like today still play a significant part in the electricity mix until 2040. Regardless of effects from climate change water will be a scarce resource in Rwanda and a thorough water management is needed.

A limiting factor of the results was time. More time could mean a more elaborated model where e.g. water storage could be integrated and necessary priorities regarding water demand revealed. More time would also mean a more thorough data collection and therefore more accurate results. Other factors that affect the outcome of the study are assumptions made and the exclusion of land use. The condition and type of soil and how land is being used are factors that have great impact on water availability. For example, overexploited land reduces the ability for water to run off and therefore causes floods.

Since land use isn't regarded that could be a possible extension of the study. Factors handled in this study (energy sources used for the electricity mix, access to electricity, climate change) all have multiple and complex social, economic and environmental effects for a country. Investigating those effects is another suggestion for future work. An energy model where heating and cooking are included could also be an interesting continuation of the study and would be a more complete analysis of the energy sector.

Another interesting improvement of this study would be an implementation of water efficient food production into OSeMOSYS which wasn't achieved this time. Instead of using the water demand as the final step in the model there would be a food demand connected to a food production. The composition of the food production would depend on water supply and the different water demands for different crops, livestock and fishing. The model would therefore find the most water efficient composition of the country's food production as an output.

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Appendix A

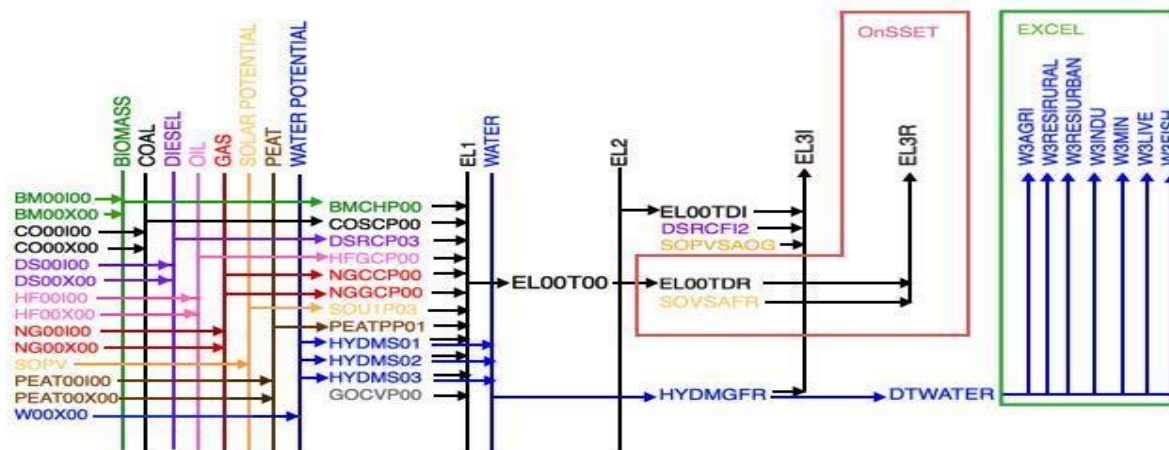


Figure 19. Rwanda's Energy and Water System. The left side shows the supply whereas the right side shows the demand and the lines represent the different fuels. Table 2 and 3 below explains the terms used in figure 1.

Table 2. Terms used for technologies in model

Technologies	
BM00I00	Biomass import
BM00X00	Biomass extraction
CO00I00	Coal import
CO00X00	Coal extraction
DS00I00	Diesel import
DS00X00	Diesel extraction
HF00I00	Oil import
HF00X00	Oil extraction
NG00I00	Natural gas import
NG00X00	Natural gas extraction
SOPV	Solar potential
PEAT00I00	Peat import
PEAT00X00	Peat extraction

W00X00	Water potential
BMCHP00	Biomass power plant
COSCP00	Coal power plant
DSRCP03	Diesel power plant
HFGCP00	Oil power plant
HYDMS01	Small hydropower plant (<1MW)
HYDMS02	Medium hydropower plant (1-5 MW)
HYDMS03	Large hydropower plant (>5 MW)
NGCCP00	Natural gas power plant combined cycle
NGGCP00	Natural gas power plant single cycle
SOU1P03	Solar power plant
PEATPP01	Peat power plant
GOCVP00	Geothermal power plant
HYDMGFR	Hydropower off grid
EL00T00	Transmission technology
EL00TDI	Distribution technology
EL00TDR	Distribution technology OnSSET
DSRCFI2	Diesel power plant off grid
SOPVSAOG	Solar stand alone
SOVSAFR	Solar power off grid
DTWATER	Water distribution technology

Table 3. Terms used for fuels in model

Fuels	
EL1	Electricity from power plants to transmission
EL2	Electricity from transmission to distribution
EL3I	Total electricity demand (excluding OnSSET demand)
EL3R	Electricity demand OnSSET
W3AGRI	Water demand Agriculture
W3RESIRURAL	Water demand Residential Rural
W3RESIURBAN	Water demand Residential Urban

W3INDU	Water demand Industry
W3MIN	Water demand Mining and quarries
W3LIVE	Water demand Livestock
W3FISH	Water demand Fisheries

Appendix B

OSEMOSYS	
Availability factors	
Biomass power plant, natural gas power plants and peat power plant	0.93
Coal power plant	0.94
Diesel power plants and oil turbine	0.9
Geothermal power plant	0.914
Other technologies	1
Capacity factors	
Biomass power plant and peat power plant	0.53
Coal power plant	0.85
Diesel power plants and oil turbine	0.9
Geothermal power plant	0.93
Natural gas power plant	0.935
Solar power plant	0.5 (only in the day)
Solar stand alone	0.5094 (only in the day)
Other technologies	1
Capital cost (MUSD/GW)	
Biomass power plant	
2010-2012	5147.958375
2013	5130.29924667114
2014	5112.64011834228
2015	5094.98099001343
2016	5077.32186168457
2017	5059.66273335571
2018	5042.00360502685
2019	5024.34447669799
2020	5006.68534836913
2021	4989.70232874687
2022	4972.71930912461

2023	4955.73628950234
2024	4938.75326988008
2025	4921.77025025782
2026	4904.78723063556
2027	4887.80421101329
2028	4870.82119139103
2029	4853.83817176877
2030	4836.8551521465
2031	4819.87213252424
2032	4802.88911290198
2033	4785.90609327972
2034	4768.92307365745
2035-2050	4751.94005403519
Coal power plant	2528.253
Diesel power plant off grid	691.59825
Diesel power plant	780.57
Transmission technology	365
Distribution technology to industry and urban areas	840.4
Distribution technology to rural areas	4233.6
Geothermal power plant	
2010-2012	5953.5496125
2013	5933.89744023177
2014	5914.24526796353
2015	5894.5930956953
2016	5874.94092342706
2017	5855.28875115883
2018	5835.63657889059
2019	5815.98440662236
2020	5796.33223435412
2021	5762.9079733676

2022	5729.48371238108
2023	5696.05945139456
2024	5662.63519040804
2025	5629.21092942152
2026	5595.78666843499
2027	5562.36240744847
2028	5528.93814646195
2029	5495.51388547543
2030	5462.08962448891
2031	5428.66536350239
2032	5395.24110251587
2033	5361.81684152934
2034	5328.39258054282
2035-2050	5294.9683195563
Oil turbine	1488.375
Small hydro power plants (< 1 MW)	
2010-2012	6231.90054375
2013	6223.98431356779
2014	6216.06808338558
2015	6208.15185320337
2016	6200.23562302116
2017	6192.31939283895
2018	6184.40316265674
2019	6176.48693247453
2020	6168.57070229232
2021	6164.49484414036
2022	6160.4189859884
2023	6156.34312783644
2024	6152.26726968448
2025	6148.19141153252

2026	6144.11555338056
2027	6140.03969522859
2028	6135.96383707663
2029	6131.88797892467
2030	6127.81212077271
2031	6123.73626262075
2032	6119.66040446879
2033	6115.58454631683
2034	6111.50868816487
2035-2050	6107.43283001291
Medium and large hydro power plants (> 1 MW)	
2010-2012	2159.346853125
2013	2165.1205078125
2014	2170.8941625
2015	2176.6678171875
2016	2182.441471875
2017	2188.2151265625
2018	2193.98878125
2019	2199.7624359375
2020	2205.536090625
2021	2214.773938125
2022	2224.011785625
2023	2233.249633125
2024	2242.487480625
2025	2251.725328125
2026	2260.963175625
2027	2270.201023125
2028	2279.438870625
2029	2288.676718125
2030	2297.914565625

2031	2307.152413125
2032	2316.390260625
2033	2325.628108125
2034	2334.865955625
2035-2050	2344.103803125
Natural gas power plant combined cycle	1181.935125
Natural gas power plant single cycle	780.57
Solar power plant	
2010-2012	1680.60239512503
2013	1626.31770586936
2014	1572.03301661368
2015	1517.748327358
2016	1463.46363810233
2017	1409.17894884665
2018	1354.89425959097
2019	1300.6095703353
2020	1246.32488107962
2021	1230.27464969525
2022	1214.22441831088
2023	1198.17418692651
2024	1182.12395554214
2025	1166.07372415777
2026	1150.0234927734
2027	1133.97326138902
2028	1117.92303000465
2029	1101.87279862028
2030	1085.82256723591
2031	1069.77233585154
2032	1053.72210446717
2033	1037.6718730828

2034	1021.62164169843
2035-2050	1005.57141031405
Hydro off grid	5100
Peat power plant	3000
Solar stand alone	5610
Fixed cost (MUSD/GW)	
Biomass power plant and peat power plant	
2010-2012	58.6405147292358
2013	58.3156365255813
2014	57.9907583219268
2015	57.6658801182723
2016	57.3410019146179
2017	57.0161237109634
2018	56.6912455073089
2019	56.3663673036545
2020	56.0414891
2021	55.9331963733333
2022	55.8249036466667
2023	55.71661092
2024	55.6083181933333
2025	55.5000254666667
2026	55.39173274
2027	55.2834400133333
2028	55.1751472866667
2029	55.06685456
2030	54.9585618333333
2031	54.8502691066666
2032	54.74197638
2033	54.6336836533333
2034	54.5253909266666

2035-2050	54.4170982
Coal power plant	26.7481597284961
Diesel power plant on grid	8
Geothermal power plant	
2010-2012	49
2013	48.875
2014	48.75
2015	48.625
2016	48.5
2017	48.375
2018	48.25
2019	48.125
2020	48
2021	47.73333333333333
2022	47.46666666666667
2023	47.2
2024	46.93333333333333
2025	46.66666666666667
2026	46.4
2027	46.13333333333333
2028	45.86666666666667
2029	45.6
2030	45.33333333333333
2031	45.06666666666667
2032	44.8
2033	44.53333333333333
2034	44.26666666666667
2035-2050	44
Small hydro power plants (< 1 MW)	
2010-2012	40.9648967819265

2013	40.8750614841857
2014	40.7852261864448
2015	40.695390888704
2016	40.6055555909632
2017	40.5157202932224
2018	40.4258849954816
2019	40.3360496977408
2020-2050	40.2462144
Medium hydro power plants (1-5 MW)	25.9113187268676
Large hydro power plants (> 5 MW)	
2010-2020	22.1541775114718
2021	22.3010083173737
2022	22.4478391232756
2023	22.5946699291774
2024	22.7415007350793
2025	22.8883315409812
2026	23.0351623468831
2027	23.181993152785
2028	23.3288239586868
2029	23.4756547645887
2030	23.6224855704906
2031	23.7693163763925
2032	23.9161471822944
2033	24.0629779881962
2034	24.2098087940981
2035-2050	24.3566396
Natural gas power plant combined cycle	1.32305861794494
Natural gas power plant single cycle	8.32894643451028
Solar power plant	
2010-2012	25

2013	24.75
2014	24.5
2015	24.25
2016	24
2017	23.75
2018	23.5
2019	23.25
2020	23
2021	22.93333333333333
2022	22.86666666666667
2023	22.8
2024	22.73333333333333
2025	22.66666666666667
2026	22.6
2027	22.53333333333333
2028	22.46666666666667
2029	22.4
2030	22.33333333333333
2031	22.26666666666667
2032	22.2
2033	22.13333333333333
2034	22.06666666666667
2035-2050	22
Hydro off grid	40.9649
Solar stand alone	
2010-2012	33
2013	32.75
2014	32.5
2015	32.25
2016	32

2017	31.75
2018	31.5
2019	31.25
2020	31
2021	30.86666667
2022	30.73333333
2023	30.6
2024	30.46666667
2025	30.33333333
2026	30.2
2027	30.06666667
2028	29.93333333
2029	29.8
2030	29.66666667
2031	29.53333333
2032	29.4
2033	29.26666667
2034	29.13333333
2035-2050	29
1/Efficiency (input activity ratio)	
Biomass and peat fuel to power plant	2.631578
Coal fuel to power plant	2.7027
Diesel fuel to power plant	2.85714
Gas fuel to power plant combined cycle	2.083333
Gas fuel to power plant single cycle	3.33
Oil fuel to power plant	2.85714
Water to small, medium and off grid hydropower plant (< 5 MW)	1497.544027794417 (million m ³)
Water to large hydropower plant (> 5 MW)	599.017611117767 (million m ³)
Other technologies	1
Operational life (years)	

Biomass, coal and peat import and extraction	35
Biomass and peat power plant	30
Coal power plant	40
Diesel import and extraction	30
Diesel power plant off grid	20
Diesel power plant on grid	25
Transmission and distribution technologies	60
Geothermal power plant and oil turbine	25
Oil import and extraction	25
Hydropower plant	50
Natural gas import	100
Natural gas extraction	30
Natural gas power plant combined cycle	30
Natural gas power plant single cycle	25
Solar potential and power plant	100
Hydropower off grid	30
Solar stand alone	15
1-Grid losses (output activity ratio)	
Distribution technology <i>2010-2050</i>	0.745-0.945 (+0.005/year)
Other technologies	1
Residual capacity (GW)	
Diesel power plant on grid	
<i>2010-2014</i>	0.0178
<i>2015-2029</i>	0.0078
Oil turbine	
<i>2010-2033</i>	0.021
Small hydropower plants (< 1 MW)	0.0001
Medium hydropower plants (1-5 MW)	0.003
Large hydropower plants (> 5 MW)	0.0495
Natural gas power plant combined cycle	

2010-2038	0.0036
Solar power plant	
2010-2032	0.00025
Specified annual demand (PJ)	
Industry and residencies electrified before 2015	
2010	1.0332
2011	1.1772
2012	1.3644
2013	1.3968
2014	1.5784
2015	1.76
2016	1.9416
2017	2.1231
2018	2.3047
2019	2.4863
2020	2.6679
2021	2.9774
2022	3.2868
2023	3.5963
2024	3.9058
2025	4.2153
2026	4.5247
2027	4.8342
2028	5.1437
2029	5.4532
2030	5.7626
2031	6.2582
2032	6.7538
2033	7.2494
2034	7.745

2035	8.2406
2036	8.7362
2037	9.2317
2038	9.7273
2039	10.2229
2040	10.7185
2041	11.6403
2042	12.5621
2043	13.4839
2044	14.4057
2045	15.3275
2046	16.2493
2047	17.171
2048	18.0928
2049	19.0146
2050	19.9364
Residencies electrified after 2015 low demand (OnSSET)	
2016	0.015638013
2017	0.024454744
2018	0.038242359
2019	0.059803449
2020	0.093520709
2021	0.146247802
2022	0.228702497
2023	0.357645253
2024	0.559286096
2025	0.874612301
2026	1.367719817
2027	2.138841969
2028	3.344723762

2029	5.230483227
2030	8.179436249
2031	8.328119468
2032	10.18638316
2033	10.39011082
2034	10.59791304
2035	10.7419851
2036	10.94071182
2037	11.14311499
2038	11.34926262
2039	11.55922398
2040	11.6282628
2041	11.814315
2042	12.00334404
2043	12.19539755
2044	12.39052391
2045	12.4596549
2046	12.6328441
2047	12.80844064
2048	12.98647796
2049	13.16699
2050	13.2143319
Residencies electrified after 2015 high demand (OnSSET)	
2016	0.016786012
2017	0.028177019
2018	0.047297976
2019	0.079394438
2020	0.133271597
2021	0.223709858
2022	0.375519628

2023	0.630347685
2024	1.058102358
2025	1.776131851
2026	2.981416994
2027	5.00461004
2028	8.400744244
2029	14.10149908
2030	23.67079278
2031	24.10107301
2032	29.47877551
2033	30.06835102
2034	30.66971804
2035	31.08665385
2036	31.66175695
2037	32.24749945
2038	32.84407819
2039	33.45169364
2040	33.6514878
2041	34.1899116
2042	34.73695019
2043	35.29274139
2044	35.85742526
2045	36.05748615
2046	36.55868521
2047	37.06685093
2048	37.58208016
2049	38.10447107
2050	38.24147565
Specified demand profile (24 hours sum up to 1)	
Day (6 AM - 6 PM)	0.34782609

Evening (6 PM - 10 PM)	0.43478261
Night (10 PM - 6 AM)	0.2173913
Total Annual Max Capacity (GW)	
Geothermal power plant	
2018-2040	0.3
2041-2050	0.7
Small hydropower plants (< 1 MW)	0.08
Medium hydropower (1-5 MW)	0.1
Large hydropower (> 5 MW)	0.3
Natural gas power plant	0.7
Peat power plant	1.2
Total Annual Max Capacity Investment (GW)	
Biomass power plant	
2013	0.05
2021-2029	0.1
2030-2039	0.25
2040-2050	0.5
Coal power plant	
2021-2029	0.1
2030-2039	0.25
2040-2050	0.5
Diesel power plant off grid	
2010-2014	0
2015	0.014
2016-2017	0
Diesel power plant on grid	
2010-2011	0
2012	0.01
2013-2015	0
2016	0.01

2017	0
2018-2029	0.1
2030-2039	0.25
2040-2050	0.5
Oil turbine	
2020-2029	0.1
2030-2039	0.25
2040-2050	0.5
Small hydropower plants (< 1 MW)	
2010	0
2011	0.00098
2012	0.0005
2013	0
2014	0.0009
2015-2019	0
2020-2030	0.0009
2030-2050	0.01125
Medium hydropower plants (1-5 MW)	
2010	0
2011	0.0048
2012	0
2013	0.0026
2014	0.0062
2015	0
2016	0.0033
2017-2018	0
2019-2030	0.01125
2031-2039	0.05
2040-2050	0.1
Large hydro power plants (> 5 MW)	

2010-2013	0
2014	0.028
2015-2020	0
2021-2023	0.0267
2024-2029	0.049
2030	0.128
2031-2050	0.1
Natural gas power plant combined cycle	
2020-2029	0.1
2030-2039	0.25
2040-2050	0.5
Natural gas power plant single cycle	
2015	0.025995
2016-2018	0
2019	0.021
2020	0
2021	0.037
2022	0
2023	0.075
2024-2029	0.1
2030-2039	0.25
2040-2050	0.5
Solar power plant	
2010-2013	0
2014	0.0085
2015-2017	0
2018-2029	0.1
2030-2039	0.25
2040-2050	0.5
Hydropower off grid	

2010-2012	0
2013	0.0002
2014	0.0014
2015	0.0001
2016-2050	0.25
Peat power plant	
2010-2015	0
2016	0.015
2017	0
2018	0.04
2019	0
2020	0.04
2021	0
2022-2050	0.04
Solar stand alone OnSSET	
2015-2050	0.25
Other technologies and years	9999999999
Total Annual Min Capacity Investment (GW)	
Diesel power plant off grid	
2015	0.014
Diesel power plant on grid	
2012 and 2016	0.01
Small hydropower plants (< 1 MW)	
2011	0.00098
2012	0.0005
2014	0.0009
Medium hydropower plants (1-5 MW)	
2011	0.0048
2013	0.0026
2014	0.0062

2016	0.0033
2019-2023	0.01125
Large hydropower plants (> 5 MW)	
2014	0.028
2021	0.0267
2024	0.049
2030	0.128
Natural gas power plant single cycle	
2015	0.025995
2019	0.021
2021	0.037
2023	0.075
Solar power plant	
2014	0.0085
Hydropower off grid	
2013	0.0002
2014	0.0014
2015	0.0001
Peat power plant	
2016	0.015
2018	0.04
2020	0.04
Total Technology Annual Activity Lower Limit (PJ)	
Diesel power plant on grid	
2010	0.2907237
2011	0.3310732
2012	0.3466
2013	0.567
2014	0.53342
2015	0.154692

2016	0.113365469
2017	0.002548177
Oil turbine	
2010	0.2847856
2011	0.35206
2012	0.38204
2013	0.3508
2014	0.35163
2015	0.361456
2016	0.217981969
2017	0.051209294
Small hydropower plants (< 1 MW)	
2010	0.0015677
2011	0.0099773
2012	0.0171754
2013	0.0090415
2014	0.015764
2015	0.0181986
2016	0.0267285
2017	0.0040541
Medium hydropower plants (1-5 MW)	
2010	0.016749
2011	0.02940611
2012	0.059151005
2013	0.064312834
2014	0.122981044
2015	0.182066576
2016	0.212734944
2017	0.0370462
Large hydropower plants (> 5 MW)	

2010	0.67203828
2011	0.761244066
2012	0.89321967
2013	0.799538764
2014	0.916088843
2015	1.136348466
2016	0.979790994
2017	0.1504513
Natural gas power plant combined cycle	
2010	0.0330212
2011	0.0219968
2012	0.03177
2013	0.0358
2014	0.05622
2015	0.055372
2016	0.016946
Natural gas power plant single cycle	
2015	0.028697
2016	0.69678698
2017	0.111484163
Solar power plant	
2010	0.0011659
2011	0.0010913
2013	0.0011
2014	0.01902
2015	0.048776
2016	0.051148
2017	0.008238643
Hydropower off grid	
2013	0.00019

2014	0.004350725
2015	0.001031217
2016	0.00467699
2017	0.00160461
Peat power plant	
2016	0.019135314
2017	0.012174336
Solar stand alone low demand (OnSSET)	
2016	0.015282023
2017	0.023589922
2018	0.036414315
2019	0.056210546
2020	0.086768772
2021	0.133939633
2022	0.206754398
2023	0.319154085
2024	0.49265859
2025	0.760486854
2026	1.17391692
2027	1.812103562
2028	2.797233145
2029	4.317917272
2030-2050	6.665304105
Stand alone high demand (OnSSET)	
2016	0.015175071
2017	0.023260886
2018	0.03565511
2019	0.054653415
2020	0.083774691
2021	0.128412814

2022	0.19683571
2023	0.301716748
2024	0.462482119
2025	0.708908975
2026	1.086640788
2027	1.665641491
2028	2.553154278
2029	3.913565316
2030-2050	5.998851544
Total Technology Activity Upper Limit (PJ)	
Biomass, coal and geothermal power plant and diesel power plant off grid	
2010-2016	0
Diesel power plant on grid	
2010	0.293630937
2011	0.334383932
2012	0.350066
2013	0.57267
2014	0.5387542
2015	0.15623892
2016	0.11449912369
Oil turbine	
2010	0.287633456
2011	0.3555806
2012	0.3858604
2013	0.354308
2014	0.3551463
2015	0.36507056
2016	0.22016178869
Small hydropower plants (< 1 MW)	
2010	0.0015677

2011	0.0099773
2012	0.0171754
2013	0.0090415
2014	0.015764
2015	0.0181986
2016	0.0267285
Medium hydropower plants (1-5 MW)	
2010	0.016749
2011	0.02940611
2012	0.059151005
2013	0.064312834
2014	0.122981044
2015	0.182066576
2016	0.212734944
Large hydropower plants (> 5 MW)	
2010	0.67203828
2011	0.761244066
2012	0.89321967
2013	0.799538764
2014	0.916088843
2015	1.136348466
2016	0.979790994
Natural gas power plant combined cycle	
2010	0.033351412
2011	0.022216768
2012	0.0320877
2013	0.036158
2014	0.0567822
2015	0.05592572
2016	0.01711546

Natural gas power plant single cycle	
2010-2014	0
2015	0.02898397
2016	0.7037548498
Solar power plant	
2010	0.001177559
2011	0.001102213
2012	0
2013	0.001111
2014	0.0192102
2015	0.04926376
2016	0.05165948
Hydropower off grid	
2010-2012	0
2013	0.0001919
2014	0.0764267
2015	0.00104152917
2016	0.0047237599
Peat power plant	
2010-2015	0
2016	0.01932666714
Solar stand alone low demand (OnSSET)	
2016	0.015590751
2017	0.024066486
2018	0.037149958
2019	0.057346112
2020	0.088521677
2021	0.136645484
2022	0.210931254
2023	0.325601642

2024	0.502611288
2025	0.775850224
2026	1.197632414
2027	1.848711715
2028	2.853742906
2029	4.405147924
2030	6.799956714
Solar stand alone high demand (OnSSET)	
2016	0.015481638
2017	0.023730803
2018	0.036375415
2019	0.055757525
2020	0.085467109
2021	0.131007012
2022	0.200812189
2023	0.307812036
2024	0.471825192
2025	0.723230368
2026	1.108593127
2027	1.699290814
2028	2.604733152
2029	3.992627241
2030	6.120040464
Water supply rcp 2.6 (million m ³)	
2010	6008.908217
2011	6337.827266
2012	5804.032083
2013	3938.323707
2014	4447.906817
2015	1814.474719

2016	7017.997223
2017	6332.433367
2018	8446.513568
2019	4361.231075
2020	9251.946541
2021	5836.631016
2022	1620.783738
2023	12428.82084
2024	3729.49327
2025	7302.247757
2026	3136.474118
2027	3542.393059
2028	4052.625069
2029	5748.15116
2030	5882.570906
2031	8885.722007
2032	6499.436028
2033	5501.132417
2034	6848.429403
2035	3467.210097
2036	2164.459871
2037	7010.215203
2038	8089.752596
2039	3123.933477
2040	7012.417468
2041	5573.832725
2042	6259.09617
2043	3665.910443
2044	8756.278782
2045	7766.40529

2046	7306.42637
2047	8508.922765
2048	2936.947024
2049	4796.107894
2050	8644.271799
Water supply rcp 6.0 (million m ³)	
2010	5792.232472
2011	6556.821441
2012	7079.083794
2013	3721.983629
2014	6414.078921
2015	7050.736366
2016	3245.146859
2017	5349.611284
2018	3113.033518
2019	8267.564692
2020	4476.653199
2021	5386.82224
2022	9283.761988
2023	6963.128919
2024	6809.346243
2025	6426.654807
2026	10162.99521
2027	4186.898218
2028	7448.18192
2029	7766.507837
2030	4486.350686
2031	6860.554262
2032	7779.896854
2033	6522.993595

2034	5559.15952
2035	4021.53303
2036	8107.64869
2037	6398.866558
2038	4657.210712
2039	3085.896573
2040	8254.030671
2041	4226.736028
2042	6375.242506
2043	5394.493713
2044	5768.889606
2045	8888.737406
2046	7847.062269
2047	6349.931306
2048	4205.82134
2049	7659.776193
2050	10972.4674
Water supply rcp 8.5 (million m ³)	
2010	3852.129432
2011	5675.990807
2012	5970.995495
2013	7276.716247
2014	14720.42875
2015	2987.295451
2016	5852.927282
2017	7630.869593
2018	6769.736755
2019	4858.738993
2020	7150.191481
2021	6076.588744

2022	6973.500942
2023	3864.696506
2024	4514.661262
2025	5230.866941
2026	8608.783251
2027	2586.053409
2028	4339.318198
2029	9658.4559
2030	1429.069911
2031	4588.052933
2032	2503.54746
2033	932.3711133
2034	6691.552021
2035	7678.465396
2036	3569.33693
2037	5469.258485
2038	5057.109284
2039	2936.894949
2040	5012.380673
2041	4590.644535
2042	9797.749631
2043	1939.221011
2044	7873.154561
2045	10090.83004
2046	5203.009805
2047	3112.042539
2048	6448.659012
2049	4635.450852
2050	5816.543888
Variable cost (MUSD/PJ)	

Biomass and peat import	1.65
Biomass and peat extraction	1.5
Biomass and peat power plant	
2010-2012	0.811261240269536
2013	0.806766718221955
2014	0.802272196174375
2015	0.797777674126794
2016	0.793283152079213
2017	0.788788630031632
2018	0.784294107984051
2019	0.77979958593647
2020	0.775305063888889
2021	0.773806889814815
2022	0.772308715740741
2023	0.770810541666667
2024	0.769312367592593
2025	0.767814193518519
2026	0.766316019444445
2027	0.76481784537037
2028	0.763319671296296
2029	0.761821497222222
2030	0.760323323148148
2031	0.758825149074074
2032	0.757326975
2033	0.755828800925926
2034	0.754330626851852
2035-2050	0.752832452777778
Coal import	
2010	3.53543859649123
2011	4.12210526315789

2012	3.33473684210526
2013	2.9140350877193
2014	2.63614035087719
2015	2.068
2016	1.76768382352941
2017	1.80189705882353
2018	1.84371323529412
2019	1.88172794117647
2020	1.92354411764706
2021	1.96536029411765
2022	2.00717647058824
2023	2.04899264705882
2024	2.09080882352941
2025	2.132625
2026	2.15337486486487
2027	2.17412472972973
2028	2.19487459459459
2029	2.21562445945946
2030	2.23637432432432
2031	2.24790202702703
2032	2.25942972972973
2033	2.27095743243243
2034	2.28248513513514
2035	2.29401283783784
2036	2.30554054054054
2037	2.31706824324324
2038	2.32859594594595
2039	2.34012364864865
2040-2050	2.35165135135135
Coal extraction	

2010	2.14269005847953
2011	2.49824561403509
2012	2.02105263157895
2013	1.76608187134503
2014	1.59766081871345
2015	1.25333333333333
2016	1.07132352941176
2017	1.09205882352941
2018	1.11740196078431
2019	1.14044117647059
2020	1.16578431372549
2021	1.19112745098039
2022	1.21647058823529
2023	1.2418137254902
2024	1.2671568627451
2025	1.2925
2026	1.30507567567568
2027	1.31765135135135
2028	1.33022702702703
2029	1.3428027027027
2030	1.35537837837838
2031	1.36236486486486
2032	1.36935135135135
2033	1.37633783783784
2034	1.38332432432432
2035	1.39031081081081
2036	1.3972972972973
2037	1.40428378378378
2038	1.41127027027027
2039	1.41825675675676

2040-2050	1.42524324324324
Coal power plant	1.16094443266042
Diesel import	
2010	19.0302336983167
2011	23.0049027618892
2012	23.5107697336166
2013	23.6312142506946
2014	21.896813204772
2015	11.5626736394836
2016	8.28658277496323
2017	10.5750285994444
2018	11.1290733780029
2019	11.7072070599771
2020	12.3335185487825
2021	12.9839189410034
2022	13.6824971400556
2023	14.4051642425233
2024	15.1519202484066
2025	15.9709429645367
2026	16.3542455956856
2027	16.7375482268345
2028	17.1208508579833
2029	17.5041534891322
2030	17.8874561202811
2031	18.27075875143
2032	18.6540613825789
2033	19.0373640137277
2034	19.4206666448766
2035	19.8039692760255
2036	20.1872719071744

2037	20.5705745383233
2038	20.9538771694721
2039	21.337179800621
2040-2050	21.7204824317699
Diesel extraction	
2010	16.4352018303644
2011	19.8678705670861
2012	20.3047556790325
2013	20.4087759437817
2014	18.910884131394
2015	9.98594541591764
2016	7.15659421474097
2017	9.13297924497467
2018	9.61147246282072
2019	10.1107697336166
2020	10.6516751103121
2021	11.2133845399575
2022	11.8167020755025
2023	12.4408236639974
2024	13.0857493054421
2025	13.7930871057362
2026	14.1241211962739
2027	14.4551552868116
2028	14.7861893773492
2029	15.1172234678869
2030	15.4482575584246
2031	15.7792916489622
2032	16.1103257394999
2033	16.4413598300376
2034	16.7723939205753

2035	17.1034280111129
2036	17.4344621016506
2037	17.7654961921883
2038	18.0965302827259
2039	18.4275643732636
2040-2050	18.7585984638013
Diesel power plant off grid	15.3888888888889
Diesel power plant on grid	0.555555555555556
Transmission and distribution technology	0.00001
Oil import	
2010	11.2193168818434
2011	13.5625919267854
2012	13.8608269325053
2013	13.9318352672005
2014	12.9093152475895
2015	6.81680013074032
2016	4.88537342703056
2017	6.23453178623958
2018	6.56117012583756
2019	6.90201013237457
2020	7.27125347278967
2021	7.65469848014381
2022	8.06654682137604
2023	8.49259682954731
2024	8.93284850465762
2025	9.41570518058506
2026	9.6416821049191
2027	9.86765902925315
2028	10.0936359535872
2029	10.3196128779212

2030	10.5455898022553
2031	10.7715667265893
2032	10.9975436509234
2033	11.2235205752574
2034	11.4494974995914
2035	11.6754744239255
2036	11.9014513482595
2037	12.1274282725936
2038	12.3534051969276
2039	12.5793821212616
2040-2050	12.8053590455957
Oil extraction	
2010	9.68941003431933
2011	11.7131475731329
2012	11.9707141689819
2013	12.0320395489459
2014	11.1489540774636
2015	5.88723647654846
2016	4.21918614152639
2017	5.38436836084327
2018	5.66646510867789
2019	5.96082693250531
2020	6.27971890831835
2021	6.6108759601242
2022	6.96656316391567
2023	7.33451544369995
2024	7.71473279947704
2025	8.13174538323255
2026	8.32690727243013
2027	8.52206916162772

2028	8.7172310508253
2029	8.91239294002288
2030	9.10755482922046
2031	9.30271671841804
2032	9.49787860761563
2033	9.6930404968132
2034	9.88820238601079
2035	10.0833642752084
2036	10.278526164406
2037	10.4736880536035
2038	10.6688499428011
2039	10.8640118319987
2040-2050	11.0591737211963
Oil turbine	4.16666666666667
Small hydropower plants (< 1 MW)	
2010-2012	2.03387914993323
2013	2.02941888848324
2014	2.02495862703325
2015	2.02049836558327
2016	2.01603810413328
2017	2.01157784268329
2018	2.00711758123331
2019	2.00265731978332
2020-2050	1.99819705833333
Medium hydropower plants (1-5 MW)	1.28647927972082
Large hydropower plants (> 5 MW)	
2010-2020	1.0999397841613
2021	1.10722983336536
2022	1.11451988256942
2023	1.12180993177348

2024	1.12909998097754
2025	1.13639003018161
2026	1.14368007938567
2027	1.15097012858973
2028	1.15826017779379
2029	1.16555022699785
2030	1.17284027620192
2031	1.18013032540598
2032	1.18742037461004
2033	1.1947104238141
2034	1.20200047301816
2035-2050	1.20929052222222
Natural gas import	
2010	9.6326599105
2011	11.2574459195
2012	12.4180073545
2013	12.8822319285
2014	11.0253336325
2015	8.0078739015
2016	6.499144036
2017	6.6152001795
2018	6.731256323
2019	6.731256323
2020	6.8473124665
2021	6.96336861
2022	7.0794247535
2023	7.195480897
2024	7.3115370405
2025	7.427593184
2026	7.7287118265946

2027	8.02983046918919
2028	8.33094911178379
2029	8.63206775437838
2030	8.93318639697297
2031	9.18411859913514
2032	9.4350508012973
2033	9.68598300345946
2034	9.93691520562162
2035	10.1878474077838
2036	10.4387796099459
2037	10.6897118121081
2038	10.9406440142703
2039	11.1915762164324
2040-2050	11.4425084185946
Natural gas extraction	
2010	9.151026914975
2011	10.694573623525
2012	11.797106986775
2013	12.238120332075
2014	10.474066950875
2015	7.607480206425
2016	6.1741868342
2017	6.284440170525
2018	6.39469350685
2019	6.39469350685
2020	6.504946843175
2021	6.6152001795
2022	6.725453515825
2023	6.83570685215
2024	6.945960188475

2025	7.0562135248
2026	7.34227623526487
2027	7.62833894572973
2028	7.9144016561946
2029	8.20046436665946
2030	8.48652707712433
2031	8.72491266917838
2032	8.96329826123243
2033	9.20168385328649
2034	9.44006944534054
2035	9.67845503739459
2036	9.91684062944865
2037	10.1552262215027
2038	10.3936118135568
2039	10.6319974056108
2040-2050	10.8703829976649
Natural gas power plant combined cycle	0.659515521121719
Natural gas power plant single cycle	0.55762782106432
Hydropower off-grid	
2010-2012	2.03387914993323
2013	2.02941888848324
2014	2.02495862703325
2015	2.02049836558327
2016	2.01603810413328
2017	2.01157784268329
2018	2.00711758123331
2019	2.00265731978332
2020-2050	1.99819705833333
ONSSET	
END_YEAR	2030

pop_2015	1129745
pop_2030	1571265
urban_ratio_2015 %	0.18467796
urban_ratio_2030 %	0.28649992
number people per household (2030)	3.2
kWh/household/year low demand	554.4
kWh/household/year high demand	1604.4
access to electricity	0.23
min_night_lights	23
pop_cutoff	770
max_grid_dist	5
max_road_dist	2
grid_price (USD/kWh)	0.003988086
grid_capacity_investment_cost (USD/kW)	1881.7
grid_losses	0.22
base_to_peak	0.416667
diesel_price	0.64395
sa_diesel_capital_cost (USD/kW)	1500
sa_pv_capital_cost (USD/kW)	5000
mg_diesel_capital_cost (USD/kW)	1000
mg_pv_capital_cost (USD/kW)	4000
mg_wind_capital_cost (USD/kW)	3000
mg_hydro_capital_cost (USD/kW)	5000
WATER DEMAND	
Urban (l/person/day)	
2012	60
2020	70
2030	80
2040	100
Rural (l/person/day)	
2012	40
2020	60
2030	80
2040	100
Industry low demand (l/capita/day)	
2012	2
2020	3
2030	5
2040	8
Industry high demand (l/capita/day)	

2012	2
2020	5
2030	10
2040	20
Mining and quarries low demand (l/capita/day)	
2012	1
2020	2
2030	3
2040	4
Mining and quarries high demand (l/capita/day)	
2012	1
2020	3
2030	5
2040	10
Livestock (l/head/day)	
50 l (%)	
2012	10
2020	25
2030	40
2040	55
20 l (%)	
2012	90
2020	75
2030	60
2040	45
Fisheries (1000 m ³ /month/ha)	
January	2
February	1
March - April	0.5
May - December	1
ha fisheries	
2012	104
2020	1100
2030	2200
2040	4400
Irrigation (m ³ /ha)	
January	1000
February	500
March - April	0
May	500
June - August	1000
September - December	500

Irrigated area (ha)	
2017	45385
2020	120000
2050	589713

Appendix C

● = Demand can't be met

Year	Month	RCP low 2.6	RCP high 2.6	RCP low 6.0	RCP high 6.0	RCP low 8.5	RCP high 8.5
TOTAL ●		198	203	197	201	207	213
2025	<i>January</i>						
	<i>February</i>						
	<i>March</i>						
	<i>April</i>						
	<i>May</i>						
	<i>June</i>						
	<i>July</i>						
	<i>August</i>						
	<i>September</i>						
	<i>October</i>		●				
	<i>November</i>						
	<i>December</i>						
2026	<i>January</i>						
	<i>February</i>						
	<i>March</i>						
	<i>April</i>						
	<i>May</i>						
	<i>June</i>	●	●				
	<i>July</i>	●	●				
	<i>August</i>	●	●				
	<i>September</i>	●	●				
	<i>October</i>	●	●				
	<i>November</i>		●				
	<i>December</i>						

2027	January						
	February						
	March						
	April	●	●				
	May	●	●				
	June	●	●				
	July	●	●		●		
	August	●	●	●	●		●
	September	●	●	●	●		●
	October	●	●	●	●	●	●
	November	●	●				
	December						
2028	January						
	February					●	●
	March					●	●
	April					●	●
	May	●	●			●	●
	June	●	●			●	●
	July	●	●	●	●	●	●
	August	●	●	●	●	●	●
	September	●	●	●	●	●	●
	October	●	●	●	●	●	●
	November	●	●	●	●	●	●
	December						
2029	January						
	February	●	●				
	March						
	April			●	●		
	May	●	●	●	●	●	●

	<i>June</i>	●	●	●	●	●	●
	<i>July</i>	●	●	●	●	●	●
	<i>August</i>	●	●	●	●	●	●
	<i>September</i>	●	●	●	●	●	●
	<i>October</i>	●	●	●	●		
	<i>November</i>						
	<i>December</i>						
2030	<i>January</i>						
	<i>February</i>						
	<i>March</i>	●	●				
	<i>April</i>	●	●				
	<i>May</i>	●	●	●	●		
	<i>June</i>	●	●	●	●		
	<i>July</i>	●	●	●	●	●	●
	<i>August</i>	●	●	●	●	●	●
	<i>September</i>	●	●	●	●	●	●
	<i>October</i>	●	●	●	●	●	●
	<i>November</i>			●	●	●	●
	<i>December</i>					●	●
2031	<i>January</i>					●	●
	<i>February</i>						
	<i>March</i>						
	<i>April</i>					●	●
	<i>May</i>	●	●			●	●
	<i>June</i>	●	●	●	●	●	●
	<i>July</i>	●	●	●	●	●	●
	<i>August</i>	●	●	●	●	●	●
	<i>September</i>	●	●	●	●	●	●
	<i>October</i>	●	●	●	●	●	●

	<i>November</i>					●	●
	<i>December</i>						
2032	<i>January</i>					●	●
	<i>February</i>			●	●	●	●
	<i>March</i>			●	●	●	●
	<i>April</i>			●	●	●	●
	<i>May</i>			●	●	●	●
	<i>June</i>			●	●	●	●
	<i>July</i>	●	●	●	●	●	●
	<i>August</i>	●	●	●	●	●	●
	<i>September</i>	●	●	●	●	●	●
	<i>October</i>	●	●	●	●	●	●
	<i>November</i>					●	●
	<i>December</i>						
2033	<i>January</i>					●	●
	<i>February</i>					●	●
	<i>March</i>					●	●
	<i>April</i>					●	●
	<i>May</i>	●	●			●	●
	<i>June</i>	●	●			●	●
	<i>July</i>	●	●			●	●
	<i>August</i>	●	●	●	●	●	●
	<i>September</i>	●	●	●	●	●	●
	<i>October</i>	●	●	●	●	●	●
	<i>November</i>						
	<i>December</i>						
2034	<i>January</i>						
	<i>February</i>						
	<i>March</i>	●	●	●	●	●	●

	<i>April</i>	●	●	●	●	●	●
	<i>May</i>	●	●	●	●	●	●
	<i>June</i>	●	●	●	●	●	●
	<i>July</i>	●	●	●	●	●	●
	<i>August</i>	●	●	●	●	●	●
	<i>September</i>	●	●	●	●	●	●
	<i>October</i>	●	●	●	●	●	●
	<i>November</i>						
	<i>December</i>						
2035	<i>January</i>						
	<i>February</i>						
	<i>March</i>	●	●				
	<i>April</i>	●	●	●	●		
	<i>May</i>	●	●	●	●	●	●
	<i>June</i>	●	●	●	●	●	●
	<i>July</i>	●	●	●	●	●	●
	<i>August</i>	●	●	●	●	●	●
	<i>September</i>	●	●	●	●	●	●
	<i>October</i>	●	●	●	●	●	●
	<i>November</i>	●	●	●	●		
	<i>December</i>			●	●		
2036	<i>January</i>						
	<i>February</i>						
	<i>March</i>	●	●	●	●		
	<i>April</i>	●	●	●	●		
	<i>May</i>	●	●	●	●		
	<i>June</i>	●	●	●	●	●	●
	<i>July</i>	●	●	●	●	●	●
	<i>August</i>	●	●	●	●	●	●

	<i>September</i>	●	●	●	●	●	●
	<i>October</i>	●	●			●	●
	<i>November</i>	●	●				
	<i>December</i>	●	●			●	●
2037	<i>January</i>	●	●	●	●	●	●
	<i>February</i>	●	●	●	●		
	<i>March</i>	●	●			●	●
	<i>April</i>	●	●	●	●	●	●
	<i>May</i>	●	●	●	●	●	●
	<i>June</i>	●	●	●	●	●	●
	<i>July</i>	●	●	●	●	●	●
	<i>August</i>	●	●	●	●	●	●
	<i>September</i>	●	●	●	●	●	●
	<i>October</i>	●	●	●	●	●	●
	<i>November</i>				●	●	●
	<i>December</i>						
2038	<i>January</i>						
	<i>February</i>			●	●		
	<i>March</i>	●	●	●	●	●	●
	<i>April</i>	●	●	●	●	●	●
	<i>May</i>	●	●	●	●	●	●
	<i>June</i>	●	●	●	●	●	●
	<i>July</i>	●	●	●	●	●	●
	<i>August</i>	●	●	●	●	●	●
	<i>September</i>	●	●	●	●	●	●
	<i>October</i>	●	●	●	●	●	●
	<i>November</i>					●	●
	<i>December</i>						

2039	January					●	●
	February				●	●	●
	March			●	●	●	●
	April			●	●	●	●
	May			●	●	●	●
	June	●	●	●	●	●	●
	July	●	●	●	●	●	●
	August	●	●	●	●	●	●
	September	●	●	●	●	●	●
	October	●	●	●	●	●	●
	November	●	●	●	●	●	●
	December	●	●	●	●	●	●
2040	January						
	February	●	●	●	●		●
	March	●	●	●	●	●	●
	April	●	●	●	●	●	●
	May	●	●	●	●	●	●
	June	●	●	●	●	●	●
	July	●	●	●	●	●	●
	August	●	●	●	●	●	●
	September	●	●	●	●	●	●
	October	●	●	●	●	●	●
	November					●	●
	December					●	●